TM Forum Specification

Resource Ordering User Guide

TMF652

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Introduction

The following document is the user guide of the REST API for Resource Order Management (TMF652). It includes the model definition as well as all available operations.

Purpose

This user guide provides comprehensive documentation for developers and system integrators implementing the TMF652 Resource Order Management API. The guide covers:

- Best practices and usage patterns
- API resource models and data structures
- Available operations and HTTP methods
- Request and response examples
- Event notifications and subscriptions

What is Resource Order Management?

The Resource Order Management API (TMF652) provides a standardized mechanism for managing the lifecycle of resource orders in telecommunications and service provider environments. A resource order represents a request to provision, modify, or delete resources that are required to deliver services to customers.

Resource Types

Resources managed through this API can be categorized as:

- **Logical Resources**: Virtual or logical entities such as:
  - MSISDN (Mobile Station International Subscriber Directory Number)
  - IP addresses and address blocks
  - VLANs (Virtual Local Area Networks)
  - Bandwidth allocations
  - Service identifiers
- **Physical Resources**: Tangible infrastructure components such as:
  - Network cables and fiber optics
  - Routers and switches
  - Firewalls and security appliances

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- Servers and storage devices
- Radio equipment and antennas

Key Capabilities

The API enables the following key capabilities:

- **Order Creation**: Create new resource orders with detailed specifications
- **Order Management**: Update, query, and track resource orders throughout their lifecycle
- **Order Cancellation**: Request and manage cancellation of resource orders
- **Item Management**: Manage individual items within a resource order, including dependencies and relationships
- **Event Notifications**: Subscribe to and receive notifications about order state changes and updates
- **Bulk Operations**: Retrieve and filter multiple resource orders efficiently

API Version and Base URL

- **Version**: 5.0.0
- **Base URL**: `https://serverRoot/tmf-api/resourceOrdering/v5/`

Target Audience

This guide is intended for:

- Software developers implementing resource order management systems
- System integrators connecting to TMF652-compliant platforms
- API consumers building applications that manage resource provisioning
- Technical architects designing service fulfillment solutions

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Sample Use Cases

This section describes common use cases and scenarios for the Resource Order Management API. These examples illustrate how the API can be used in real-world telecommunications and service provider environments.

Use Case 1: New Service Activation - Mobile Number Provisioning

Scenario

A customer subscribes to a new mobile service plan. The service provider needs to provision a new MSISDN (mobile number) and associated logical resources.

Steps

1. **Create Resource Order**: The order management system creates a new resource order with action type "add"
2. **Specify Resources**: Include resource order items for:
   - MSISDN allocation
   - SIM card association
   - Network profile configuration
3. **Submit Order**: POST the resource order to the API
4. **Monitor Progress**: Subscribe to state change notifications to track provisioning
5. **Completion**: Receive notification when resources are successfully provisioned

Key Attributes

- `action`: "add"
- `resourceOrderItem.resource.resourceSpecification`: Reference to MSISDN specification
- `state`: Transitions from "acknowledged" → "inProgress" → "completed"

Use Case 2: Service Upgrade - Bandwidth Increase

Scenario

An existing customer requests an upgrade to their internet service, requiring modification of bandwidth allocation resources.

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Steps

1. **Create Modify Order**: Create a resource order with action type "modify"
2. **Reference Existing Resources**: Include references to the current bandwidth allocation resources
3. **Specify New Values**: Define the new bandwidth characteristics
4. **Handle Dependencies**: Ensure proper sequencing if multiple resources need updating
5. **Track Completion**: Monitor order state until upgrade is complete

Key Attributes

- `action`: "modify"
- `resourceOrderItem.resource.id`: Reference to existing resource
- `resourceOrderItem.resource.resourceCharacteristic`: Updated bandwidth values

Use Case 3: Service Deactivation – Customer Churn

Scenario

A customer cancels their service. All associated resources (logical and physical) need to be released and returned to inventory.

Steps

1. **Create Delete Order**: Create a resource order with action type "delete"
2. **Identify Resources**: List all resources to be deprovisioned:
   - MSISDN release
   - IP address release
   - Equipment return
3. **Submit Order**: POST the deletion order
4. **Coordinate Dependencies**: Ensure proper order of resource deletion
5. **Verify Completion**: Confirm all resources are successfully released

Key Attributes

- `action`: "delete"

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- `resourceOrderItem`: List of resources to be removed
- `state`: Tracks deprovisioning progress

Use Case 4: Network Infrastructure Deployment

Scenario

Deploy new network infrastructure including physical equipment (routers, switches) and logical configurations (VLANs, routing tables).

Steps

1. **Create Complex Order**: Create a resource order with multiple items
2. **Physical Resources**: Include order items for:
   - Router installation
   - Switch deployment
   - Cable connections
3. **Logical Resources**: Include order items for:
   - VLAN configuration
   - IP address assignments
   - Routing rules
4. **Define Dependencies**: Use `resourceOrderItem.orderItemRelationship` to specify installation sequence
5. **Track Installation**: Monitor each item's state independently

Key Attributes

- Multiple `resourceOrderItem` entries
- `resourceOrderItem.orderItemRelationship`: Defines dependencies between items
- `resourceOrderItem.appointment`: Scheduling for physical installations

Use Case 5: Order Cancellation

Scenario

A resource order needs to be cancelled before completion, either due to customer request

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or business reasons.

Steps

1. **Check Order State**: Retrieve the order using GET `/resourceOrder/{id}` and verify the `state` attribute indicates the order is in a cancellable state (e.g., `acknowledged`, `inProgress`, but not `completed` or `failed`)
2. **Create Cancellation Request**: POST to `/cancelResourceOrder` endpoint
3. **Provide Reason**: Include cancellation reason and requested date
4. **Reference Original Order**: Link to the resource order being cancelled
5. **Monitor Cancellation**: Track the cancellation request state
6. **Handle Partial Completion**: Manage any resources already provisioned

Key Attributes

- `cancelResourceOrder.resourceOrder`: Reference to order being cancelled
- `cancelResourceOrder.cancellationReason`: Explanation for cancellation
- `cancelResourceOrder.requestedCancellationDate`: When cancellation should occur

Use Case 6: Bulk Order Retrieval and Monitoring

Scenario

An operations dashboard needs to display all active resource orders and their current status for monitoring purposes.

Steps

1. **Query Active Orders**: GET resource orders with state filter
2. **Apply Pagination**: Use offset and limit parameters for large result sets
3. **Filter by Date**: Use before/after parameters to focus on recent orders
4. **Sort Results**: Order by creation date or priority
5. **Subscribe to Updates**: Register webhooks for real-time notifications

Query Parameters

- `state`: "inProgress" or "acknowledged"

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- `offset`: 0
- `limit`: 50
- `sort`: "creationDate:desc"
- `after`: Recent date filter

Use Case 7: Order Modification - Adding Items to Existing Order

Scenario

After submitting a resource order, additional resources need to be added to the same order.

Steps

1. **Retrieve Current Order**: GET the existing resource order
2. **Prepare Update**: Create PATCH request with new order items
3. **Maintain Consistency**: Ensure new items are compatible with existing ones
4. **Submit Update**: PATCH the resource order
5. **Verify Changes**: Confirm the order reflects the additions

Key Attributes

- HTTP Method: PATCH
- Content-Type: `application/json` or `application/json-patch+json`
- Include new `resourceOrderItem` entries

Use Case 8: Event-Driven Integration

Scenario

An external system needs to react to resource order lifecycle events in real-time.

Steps

1. **Register Listener**: POST to `/hub` to register a notification endpoint
2. **Specify Events**: Define which event types to receive:
   - `ResourceOrderStateChangeEvent`
   - `ResourceOrderCreateEvent`

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- `ResourceOrderAttributeValueChangeEvent`

3. **Receive Notifications**: External system receives HTTP POST with event data
4. **Process Events**: Take appropriate actions based on event type and payload
5. **Maintain Subscription**: Periodically verify hub registration is active

Event Types

- `ResourceOrderCreateEvent`: New order created
- `ResourceOrderStateChangeEvent`: Order state transitions
- `ResourceOrderAttributeValueChangeEvent`: Order attributes modified
- `ResourceOrderInformationRequiredEvent`: Additional information needed
- `ResourceOrderDeleteEvent`: Order deleted

Use Case 9: Scheduled Resource Provisioning

Scenario

Resources need to be provisioned at a specific future date and time, such as for a planned service launch.

Steps

1. **Create Order with Appointment**: Include appointment details in the order
2. **Specify Timing**: Set `requestedCompletionDate` for when provisioning should occur
3. **Coordinate Resources**: Ensure all dependent resources are scheduled appropriately
4. **Monitor Schedule**: Track order until the scheduled time
5. **Execute Provisioning**: System automatically processes order at scheduled time

Key Attributes

- `requestedCompletionDate`: Target completion date/time
- `resourceOrderItem.appointment`: Detailed scheduling information
- `resourceOrderItem.appointment.id`: Appointment reference

Use Case 10: Multi-Party Order Coordination

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Scenario

A resource order involves multiple parties (customer, installer, network operations) who need to coordinate.

Steps

1. **Define Related Parties**: Include all parties in the order
2. **Specify Roles**: Assign appropriate roles to each party
3. **Reference Party Details**: Link to party management system
4. **Track Responsibilities**: Monitor which party is responsible for each order item
5. **Coordinate Communication**: Use order notes for inter-party communication

Key Attributes

- `relatedParty.role`: Party's role in the order
- `relatedParty.partyOrPartyRole.id`: Reference to party
- `relatedParty.partyOrPartyRole.@referredType`: Type of party entity
- `note`: Communication and coordination notes

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

ResourceOrder resource

A Resource Order is a request to provision a set of Resources (logical and physical) triggered by the request to provision a Service through a Service Order.

Resource model

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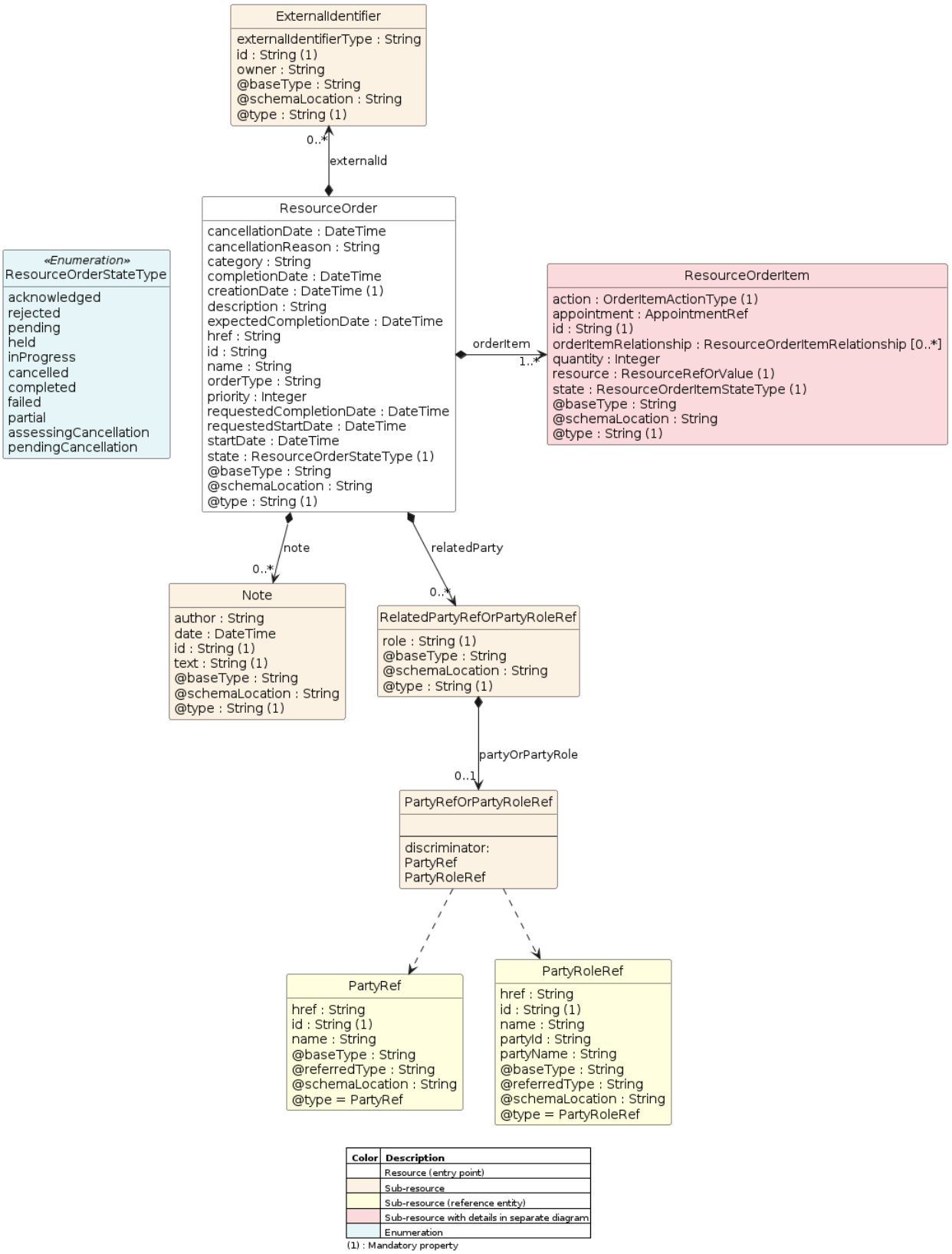


Figure 1 - ResourceOrder

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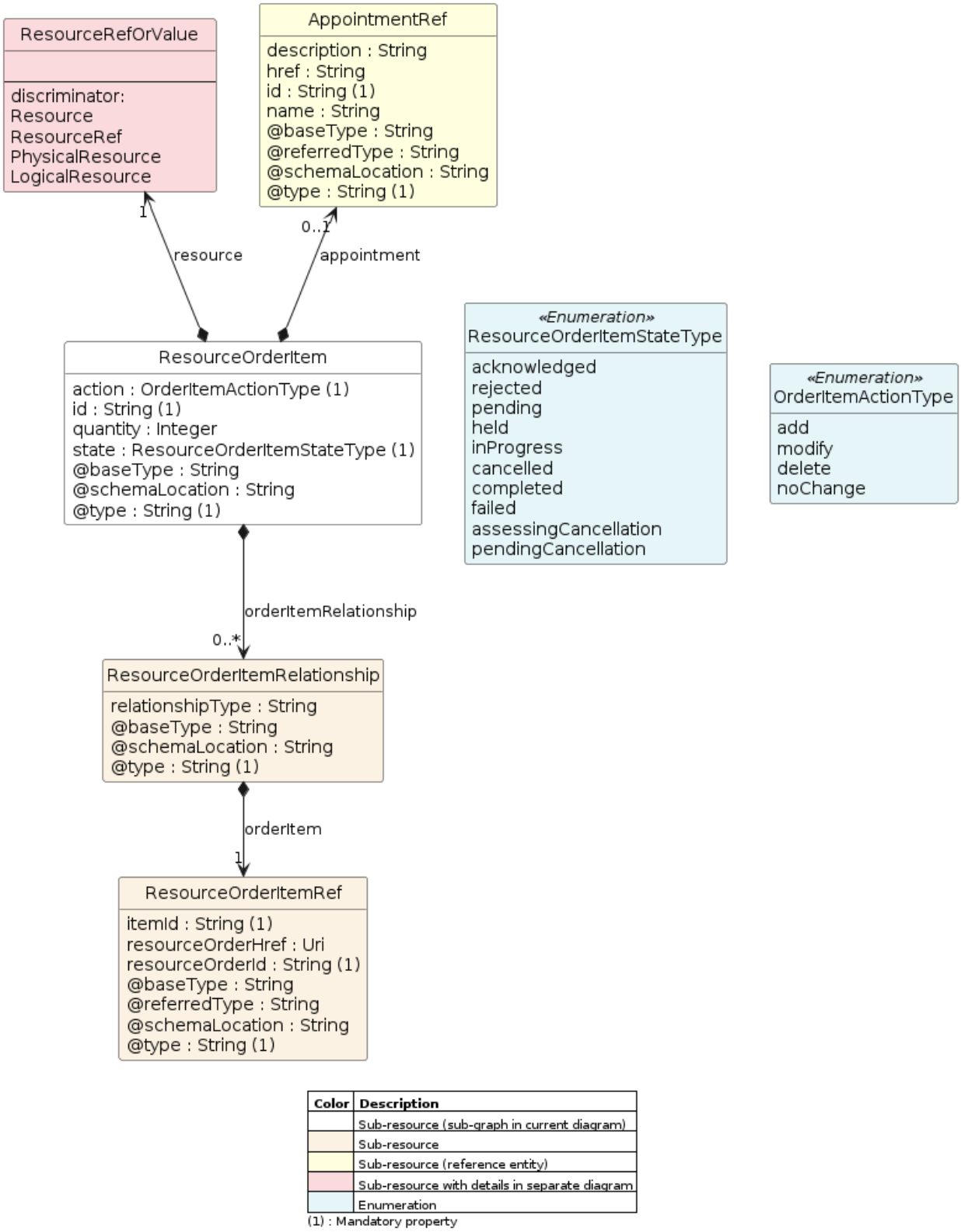


Figure 2 – ResourceOrderItem

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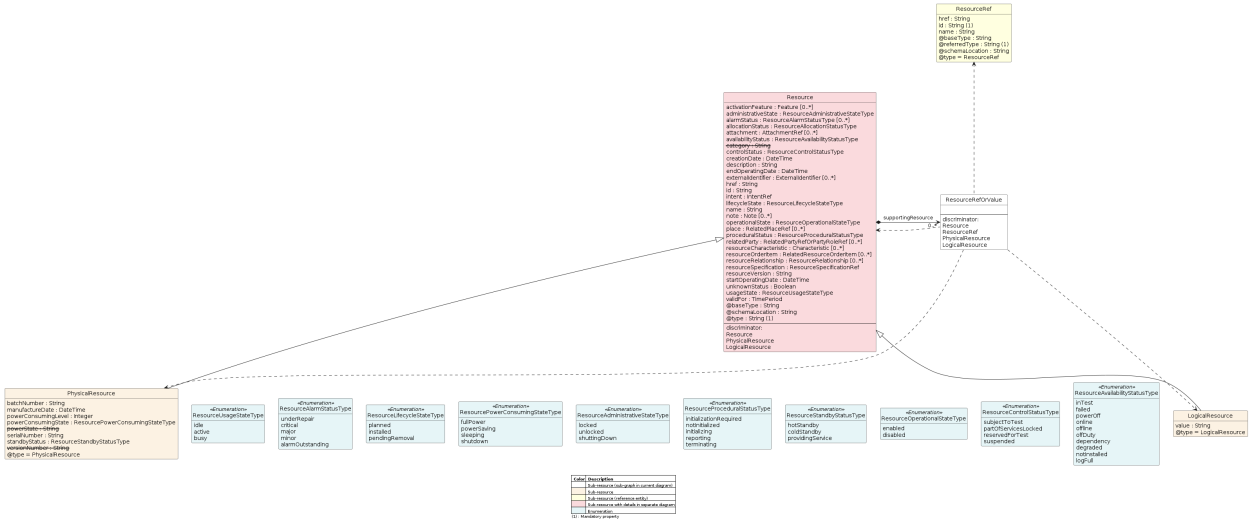


Figure 3 - ResourceRefOrValue

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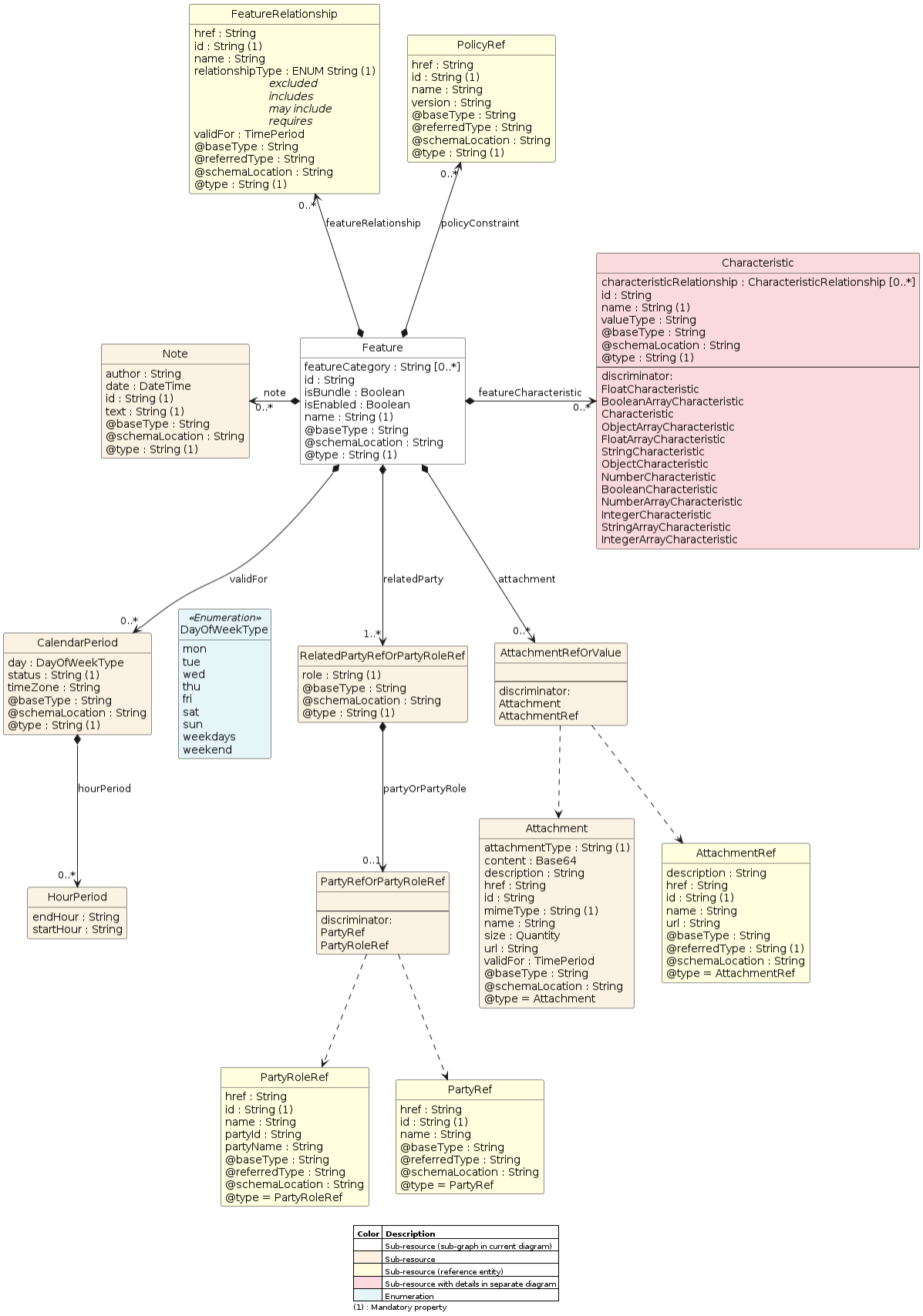


Figure 4 - Feature

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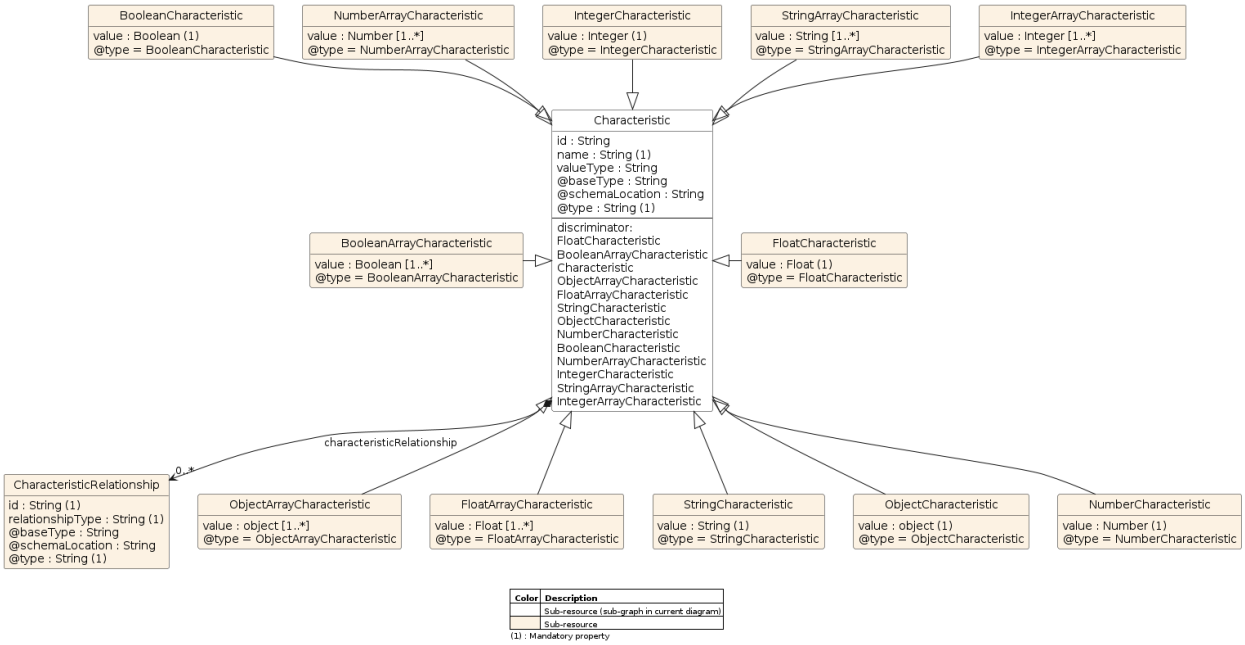


Figure 5 - Characteristic

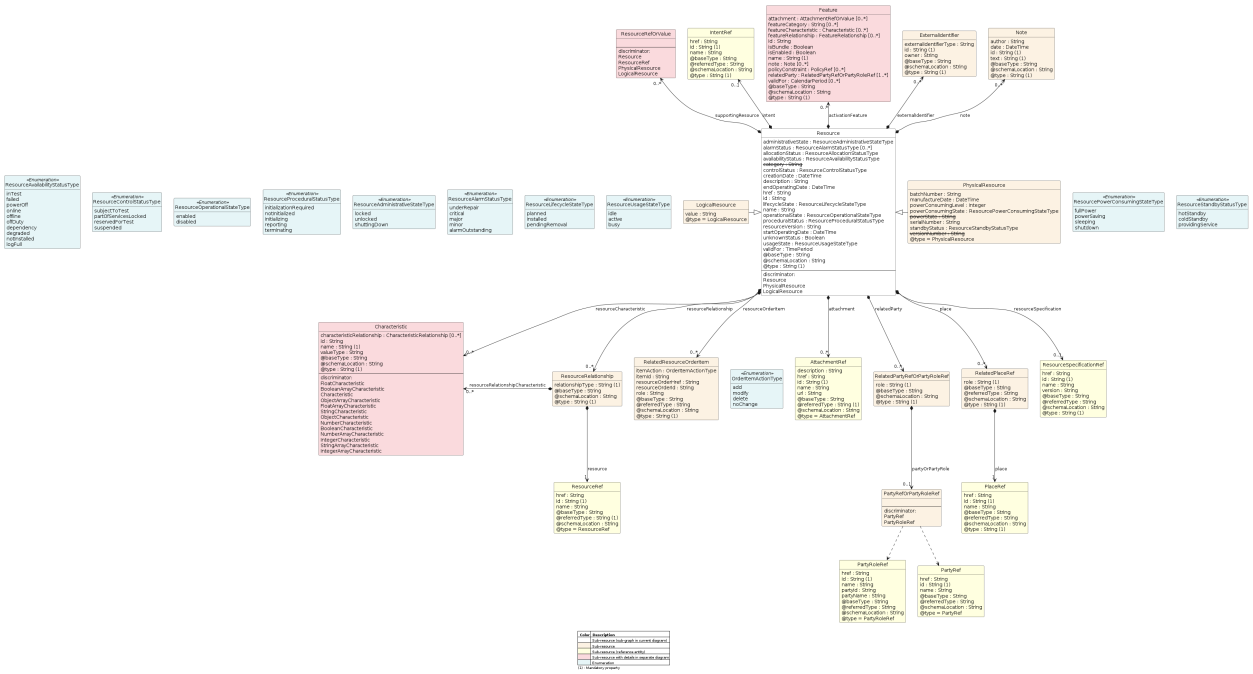


Figure 6 - Resource

Field descriptions

ResourceOrder fields

cancellationDate	A DateTime. Date when the order is cancelled - if cancelled, in RFC3339 (subset of ISO8601) format.

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cancellationReason	A String. Reason why the order is cancelled - if cancelled.
category	A String. Used to categorize the order from a business perspective that can be useful for the order management system.
completionDate	A DateTime. Date when the order was actually completed, in RFC3339 (subset of ISO8601) format.
creationDate	A DateTime. Date when the order was created, in RFC3339 (subset of ISO8601) format.
description	A String. Free-text description of the Resource Order.
expectedCompletionDate	A DateTime. Date when the order is expected to be completed, in RFC3339 (subset of ISO8601) format.
externalId	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A string used to give a name to the Resource Order.
note	A Note. A list of notes containing additional human-readable information associated with the resource order.
orderItem	A ResourceOrderItem. A list of resource order items, each representing a discrete action to be performed as part of the resource order.
orderType	A String. Name of the Resource Order type.
priority	An Integer. A way that can be used by consumers to prioritize orders in OM system (such as: 0 is the highest priority, and 4 the lowest).
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of parties which are involved in this order and the role they are playing.
requestedCompletionDate	A DateTime. Requested delivery date from the requestor perspective, in RFC3339 (subset of ISO8601) format.
requestedStartDate	A DateTime. Order start date wished by the requestor, in RFC3339 (subset of ISO8601) format.
startDate	A DateTime. Date when the order was actually started, in RFC3339 (subset of ISO8601) format.

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state	A ResourceOrderStateType. Possible values for the state of the order. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * partial * assessingCancellation * pendingCancellation
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.

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id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceOrderItem sub-resource fields

action	An OrderItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
appointment	An AppointmentRef. Refers an appointment, such as a Customer presentation or internal meeting or site visit.
id	A String. Identifier of the line item (such as a sequence number 01, 02, 03).
orderItemRelationship	A ResourceOrderItemRelationship. A list of relationships that this order item has with other order items within the same resource order. These relationships describe dependencies or associations such as parent-child, replacement, or bundled items.
quantity	An Integer. Quantity ordered.
resource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.

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state	A ResourceOrderItemStateType. Possible values for the state of the order item. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * assessingCancellation * pendingCancellation
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AppointmentRef sub-resource fields

description	A String. An explanatory text regarding the appointment made with a party.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceOrderItemRelationship sub-resource fields

orderItem	A ResourceOrderItemRef.
relationshipType	A String. Type of the order item relationship.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceOrderItemRef sub-resource fields

itemId	A String. Identifier of the line item.
resourceOrderHref	A Uri. Link to the order to which this item belongs to.
resourceOrderId	A String. Identifier of the resource order that this item belongs to.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LogicalResource sub-resource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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Feature sub-resource fields

attachment	An AttachmentRefOrValue. Attachments relevant to this Feature, such as documentation.
featureCategory	A String. Collection of feature categories.
featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. The name for the feature.
note	A Note. Notes relevant to this Feature.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Party that may be relevant to this Feature.
validFor	A CalendarPeriod. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

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id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values: * mon * tue * wed * thu * fri * sat * sun * weekdays * weekend
hourPeriod	A HourPeriod. Start and end time range in 24-hour HH:mm format.
status	A String. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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HourPeriod sub-resource fields

endHour	A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

place	A PlaceRef. A reference to a physical Place.
role	A String. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The kind of Place that is being referenced.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

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NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedResourceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRelationship sub-resource fields

relationshipType	A String. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRef. Resource reference, for when Resource is used by other entities.
resourceRelationshipCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic... Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhysicalResource sub-resource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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batchNumber	A String. A unique identifier assigned to a specific group of physical resources produced or received under the same conditions for tracking and traceability in the supply chain.
~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.

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endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the ISO 8601 date time standard format. This is an optional attribute.
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.

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operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
powerConsumingLevel	An Integer. The power consuming level attribute (defined by ITU-T M.3701) indicates the current percentage of power consuming of the resource represented by the managed object, where 100 percent represents the maximum power consumption of the device. It is single valued and read-only by management functions.

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powerConsumingState	A ResourcePowerConsumingStateType. Value enumeration for the 'resourcePowerConsumingState' of a Resource; the power consuming state attribute (defined by ITU-T M.3701) indicates the power consuming policy of the resource represented by the managed object, it is single valued and read-write by management functions.; defined values: 'fullPower': The resource runs in full power and will not consider power saving. If a large device contains parts that also support this attribute, when the container object set to full power state, all its parts shall turn into full power state; 'powerSaving': The resource is (or may be) partially running to save power; depending on how much the payload is lower than the maximum capacity of the resource, the degree of saving will be automatically decided by the resource. If power saving is not supported by the resource, attempting to change to this state will have the same effect as full power state. If a large device contains parts that also support this attribute, when the container object is set to power saving state, its parts may still remain in full power state. But if one of the parts turns into power saving state, the container object shall turn into power saving state; 'sleeping': The resource is not running but keeps a snapshot of its memory before sleeping, and can resume running mode quickly. In this mode, the power is only consumed to meet the need of maintaining the memory. If it is not supported by the resource, attempting to change to this state will result in a failure; 'shutdown': The resource has been powered off and does not consume any more power; if it is not supported by the resource, attempting to change to this state will result in a failure. ENUMERATED with values: * fullPower * powerSaving * sleeping * shutdown

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~~powerState~~	DEPRECATED A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off; NOTE: This attribute is replaced by the 'powerConsumingState' attribute, to align with the official name as defined by ITU-T.

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).

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resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and PartNumber attributes are used to identify different types of hardware items.
standbyStatus	A ResourceStandbyStatusType. Value enumeration for the 'resourceStandbyStatus' of a Resource; the standby status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the working mode for standby objects, i.e. whether it is providing service or standby as a back-up resource; it is single valued and read-only by management functions; defined values: 'hotStandby': The resource is not providing service, but is operating in synchronism with another resource that is to be backed-up (e.g., a computer shadowing another computer). A resource with a hot standby status will be immediately able to take over the role of the resource to be backed up, without the need for initialization activity, and will contain the same information as the resource to be backed up; 'coldStandby': The resource is to back up another resource, but is not synchronized with that resource. A resource with a cold standby status will not be immediately able to take over the role of a resource to be backed up, and will require some initialization activity; 'providingService': The back up resource is providing service and is backing up another resource. ENUMERATED with values: * hotStandby * coldStandby * providingService
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).

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unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
~~versionNumber~~	DEPRECATED A String. This is a string that identifies the version of this physical resource. This is an optional attribute. NOTE that this attribute is deprecated as it overlaps with the Resource.resourceVersion attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	Resource can be instantiated as * LogicalResource * PhysicalResource
batchNumber	This property is present in subclasses
manufactureDate	This property is present in subclasses
powerConsumingLevel	This property is present in subclasses
powerConsumingState	This property is present in subclasses
powerState	This property is present in subclasses
serialNumber	This property is present in subclasses
standbyStatus	This property is present in subclasses
value	This property is present in subclasses
versionNumber	This property is present in subclasses

ResourceRef sub-resource fields

See [ResourceRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ResourceOrder' resource object.


```
{
  "@type": "ResourceOrder",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "owner": "ExternalSystem",
      "externalIdentifierType": "ContractReference",
      "id": "EXT123456"
    }
  ],
  "id": "54321",
  "href": "https://host:port/tmf-api/resourceOrder/resourceOrder/54321",
  "@schemaLocation": "/schema-registry/schema/ResourceOrder",
  "creationDate": "2025-01-22T06:41:23Z",
  "name": "New Network Deployment Order",
  "description": "This order pertains to the deployment of a new network
infrastructure.",
  "priority": 1,
```


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```
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "startDate": "2025-01-23T06:41:23Z",
  "expectedCompletionDate": "2025-02-12T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T06:41:23Z",
      "text": "Order created and acknowledged by the operations team."
    }
  ],
  "relatedParty": [
    {
      "role": "member",
      "partyOrPartyRole": {
        "id": "9866",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/9866",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 2,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "https://host:port/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "InstallationAppointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
```


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```
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "https://host:port/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "@referredType": "ResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "https://host:port/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "SiteLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
```


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```
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "https://host:port/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "TelecommunicationsProvider"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "https://host:port/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "@referredType": "ResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "https://host:port/tmf-api/v5/place/98765",
```


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```
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "SiteLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "https://host:port/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "TelecommunicationsProvider"
                }
              }
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "https://host:port/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
```


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```
                "@baseType": "Characteristic",
                "value": 100,
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "https://host:port/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "TechnicalDocumentation",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "https://host:port/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "https://host:port/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "@referredType": "SecurityFeature",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
```


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```
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "value": 50,
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "12345",
                        "relationshipType": "enhances"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
                  {
                    "@type": "PolicyRef",
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "https://host:port/tmf-api/v5/policy/12345",
                    "id": "12345",
                    "name": "Service Level Agreement",
                    "@referredType": "ServicePolicy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": true,
                "id": "12345",
                "name": "Firewall Protection"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "https://host:port/tmf-api/v5/intent/12345",
              "id": "12345",
              "name": "Security Enhancement",
              "@referredType": "ServiceIntent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
```


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```
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "value": 5000,
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "@referredType": "Resource",
              "id": "87654",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "https://host:port/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "value": 2048,
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "https://host:port/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Document",
            "description": "Detailed product specification document.",
```


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```
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "https://host:port/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "https://host:port/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "@referredType": "NetworkFeature",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "value": 5000,
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "https://host:port/tmf-api/v5/policy/12345",
                "id": "12345",
```


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```
                "name": "Data Retention Policy",
                "@referredType": "DataPolicy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "https://host:port/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "ServiceUpgrade"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}
```


CancelResourceOrder resource

Request for cancellation an existing resource order.

Resource model

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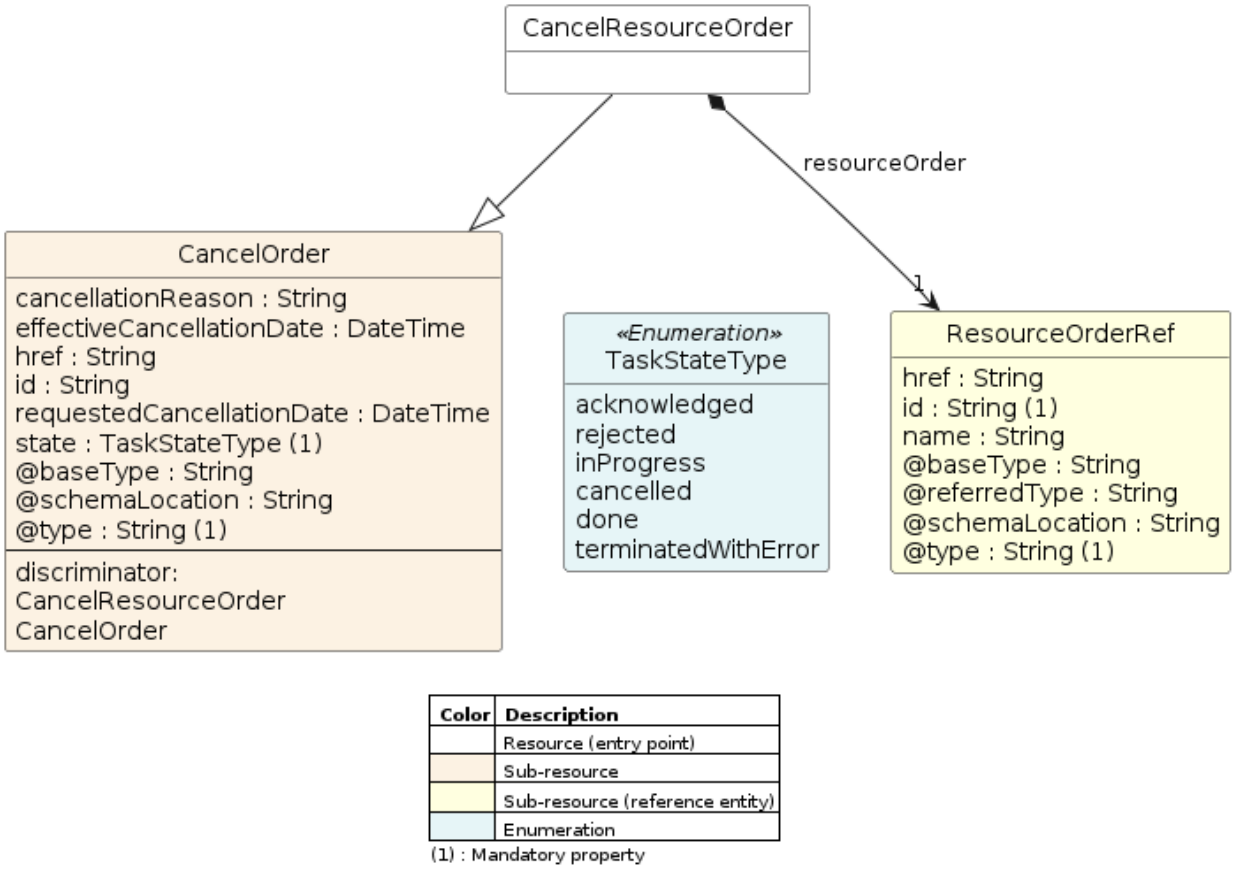


Figure 7 - CancelResourceOrder

Field descriptions

CancelResourceOrder fields

cancellationReason	A String. Reason why the order is cancelled.
effectiveCancellationDate	A DateTime. Date when the order is cancelled.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
requestedCancellationDate	A DateTime. Date when the submitter wants the order to be cancelled.
resourceOrder	A ResourceOrderRef.

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state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceOrderRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'CancelResourceOrder' resource object.

```
{
  "@type": "CancelResourceOrder",
  "id": "54321",
  "href": "https://host:port/tmf-api/resourceOrder/CancelResourceOrder/54321",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
```

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```
   "state": "acknowledged",
   "resourceOrder": {
     "@type": "ResourceOrderRef",
     "@schemaLocation": "/schema-registry/schema/ResourceOrder",
     "href": "https://host:port/tmf-
 api/resourceOrderingManagement/v5/resourceOrder/12345",
     "id": "12345",
     "name": "Network Infrastructure Upgrade",
     "@referredType": "ResourceOrder"
   }
 }
```

Notification Resource Models

8 notifications are defined for this API.

Notifications related to ResourceOrder:

- Create Event
- Delete Event
- Attribute Value Change Event
- Information Required Event
- State Change Event

Notifications related to CancelResourceOrder:

- Create Event
- Information Required Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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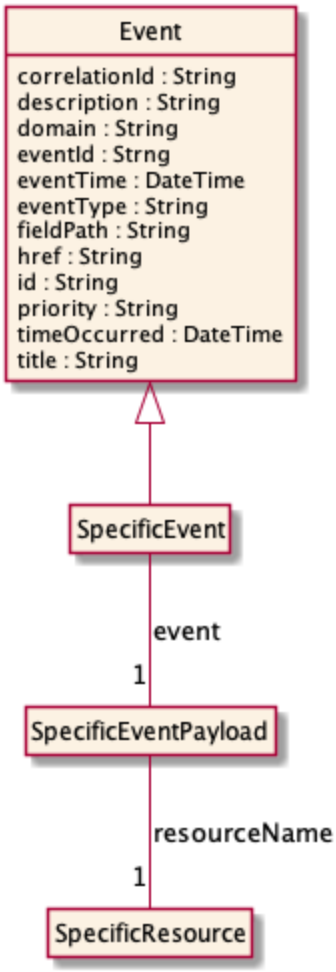


Figure 8 Notification Pattern

ResourceOrder

Create Event

Notification sent when a new resource order is created

```
Content-Type: application/json
{
  "@type": "ResourceOrderCreateEvent",
  "@schemaLocation": "/schema-registry/schema/ResourceOrderCreateEvent",
  "href": "/tmf-api/v5/events/resourceOrderCreate/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Resource Order Create",
  "description": "This event creates a resource order.",
  "priority": "medium",
  "timeOccurred": "2025-01-22T06:41:27Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
```

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```
      "href": "/tmf-api/v5/systems/sourceSystem/source123",
      "id": "source123",
      "name": "SourceSystem"
    },
    "reportingSystem": {
      "@type": "System",
      "@schemaLocation": "/schema-registry/schema/ReportingSystem",
      "href": "/tmf-api/v5/systems/reportingSystem/report567",
      "id": "report567",
      "name": "ReportingSystem"
    },
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation": "/schema-registry/schema/RelatedParty",
        "role": "Customer",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "/schema-registry/schema/Party",
          "href": "/tmf-api/v5/party/customer789",
          "id": "customer789",
          "name": "John Doe",
          "@referredType": "Individual"
        }
      }
    ],
    "analyticCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "@baseType": "Characteristic",
        "@schemaLocation": "/schema-registry/schema/Characteristic",
        "id": "char123",
        "name": "OrderPriorityChange",
        "valueType": "string",
        "value": "High",
        "characteristicRelationship": [
          {
            "@type": "CharacteristicRelationship",
            "id": "rel456",
            "relationshipType": "associated"
          }
        ]
      }
    ],
    "eventId": "event56789",
    "eventTime": "2025-01-22T06:41:27Z",
    "eventType": "Create",
    "event": {
      "resourceOrder": {
        "@type": "ResourceOrder",
        "cancellationDate": "2025-01-22T06:41:27Z",
        "cancellationReason": "Duplicate order identified",
        "externalId": [
          {
            "@type": "ExternalIdentifier",
            "id": "EXT00001",
            "owner": "ExternalSystem",
            "externalIdentifierType": "OrderReference"
          }
        ],
```


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```
      "name": "Customer Order Cancellation",
      "description": "Cancellation of a duplicate customer order to avoid redundancy.",
      "priority": 2,
      "orderType": "Cancellation",
      "category": "Customer Request",
      "requestedStartDate": "2025-01-22T06:00:00Z",
      "requestedCompletionDate": "2025-01-22T06:30:00Z",
      "startDate": "2025-01-22T06:05:00Z",
      "completionDate": "2025-01-22T06:25:00Z",
      "expectedCompletionDate": "2025-01-22T06:30:00Z",
      "note": [
        {
          "@type": "Note",
          "id": "note123",
          "author": "Jane Doe",
          "date": "2025-01-22T06:10:00Z",
          "text": "Cancellation approved by customer service team."
        }
      ],
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "@schemaLocation": "/schema-registry/schema/RelatedParty",
          "role": "Customer",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "@schemaLocation": "/schema-registry/schema/Party",
            "href": "/tmf-api/v5/party/party123",
            "id": "party123",
            "name": "Service Provider Organization",
            "@referredType": "Organization"
          }
        }
      ],
      "creationDate": "2025-01-22T06:41:27Z",
      "state": "acknowledged",
      "orderItem": [
        {
          "@type": "ResourceOrderItem",
          "orderItemRelationship": [
            {
              "@type": "ResourceOrderItemRelationship",
              "@schemaLocation": "/schema-
registry/schema/ResourceOrderItemRelationship",
              "relationshipType": "dependency",
              "orderItem": {
                "@type": "ResourceOrderItemRef",
                "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
                "itemId": "ITEM56789",
                "resourceOrderId": "ORDER12345",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
                "@referredType": "ResourceOrderItem"
              }
            }
          ],
          "quantity": 500,
          "appointment": {
            "@type": "AppointmentRef",
            "@schemaLocation": "/schema-registry/schema/AppointmentRef",
            "href": "/tmf-api/v5/appointment/67890",
```


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```
            "id": "67890",
            "name": "Site Installation Appointment",
            "@referredType": "InstallationAppointment",
            "description": "Appointment for the installation of network infrastructure
at the customer site."
          },
          "action": "add",
          "id": "12345",
          "resource": {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "category": "Network Element",
            "description": "Fiber Optic Cable connecting two locations for high-speed
data transmission.",
            "name": "Fiber Optic Cable - Main Line",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-02-25T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary",
                "@referredType": "ResourceOrderItem"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "SiteLocation"
                }
              }
```


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```
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "TelecommunicationsProvider"
                }
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "CH123456",
                "name": "Storage Capacity",
                "valueType": "integer",
                "value": 2048,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "CHREL7890",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/Attachment",
                "href": "/tmf-api/v5/attachment/ATT98765",
                "id": "ATT98765",
                "name": "Product Specification",
                "@referredType": "Document",
                "description": "Detailed product specification document.",
                "url": "http://example.com/downloads/product-spec"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
              "href": "/tmf-api/v5/specification/SPEC54321",
              "id": "SPEC54321",
              "name": "Enterprise Storage",
              "@referredType": "ResourceSpecification",
              "version": "2.1"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "3.0",
```


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```
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": false,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Network Upgrade",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "45678",
                    "name": "Speed",
                    "valueType": "Integer",
                    "value": 1000,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "34567",
                        "relationshipType": "associated"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
                  {
                    "@type": "PolicyRef",
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "/tmf-api/v5/policy/12345",
                    "id": "12345",
                    "name": "Data Retention Policy",
                    "@referredType": "DataPolicy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": false,
                "id": "23456",
                "name": "Service Bundle"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "/tmf-api/v5/intent/INT1001",
              "id": "INT1001",
```


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```
              "name": "Upgrade Service",
              "@referredType": "Intent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "XYZ Corp",
                "externalIdentifierType": "ProjectID",
                "id": "EXT45678"
              }
            ]
          },
          "state": "acknowledged"
        }
      ]
    }
  }
}
```

Delete Event

Notification sent when a resource order is deleted

```
Content-Type: application/json
{
  "@type": "ResourceOrderDeleteEvent",
  "@schemaLocation": "/schema-registry/schema/resourceOrderDeleteEvent",
  "href": "/tmf-api/v5/events/resourceOrderDeleteEvent/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Resource Order Delete",
  "description": "This event deletec a resource order.",
  "priority": "medium",
  "timeOccurred": "2025-01-22T06:41:27Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
```

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```
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "char123",
      "name": "OrderPriorityChange",
      "valueType": "string",
      "value": "High",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "rel456",
          "relationshipType": "associated"
        }
      ]
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-01-22T06:41:27Z",
  "eventType": "Delete",
  "event": {
    "resourceOrder": {
      "@type": "ResourceOrder",
      "id": "12345",
      "href": "/tmf-api/v5/resourceOrder/12345"
    }
  }
}
```


Attribute Value Change Event

Notification sent when resource order attributes are modified


```
Content-Type: application/json
{
  "@type": "ResourceOrderAttributeValueChangeEvent",
  "@schemaLocation": "/schema-registry/schema/ResourceOrderAttributeValueChangeEvent",
  "href": "/tmf-api/v5/events/attributeValueChange/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Resource Order Attribute Value Change",
  "description": "This event captures changes made to the attributes of a resource
order.",
  "priority": "medium",
  "timeOccurred": "2025-01-22T06:41:27Z",
```


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```
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "char123",
      "name": "OrderPriorityChange",
      "valueType": "string",
      "value": "High",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "rel456",
          "relationshipType": "associated"
        }
      ]
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-01-22T06:41:27Z",
  "eventType": "AttributeValueChange",
  "event": {
    "resourceOrder": {
      "@type": "ResourceOrder",
      "id": "12345",
      "href": "/tmf-api/v5/resourceOrder/12345",
      "priority": 4
    }
  }
}
```


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Information Required Event

Notification sent when additional information is needed to process a resource order

```
Content-Type: application/json
{
  "@type": "ResourceOrderInformationRequiredEvent",
  "@schemaLocation": "/schema-registry/schema/resourceOrderInformationRequiredEvent",
  "href": "/tmf-api/v5/events/informationRequired/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Resource Order Information Required",
  "description": "This event captures information required for a resource order.",
  "priority": "medium",
  "timeOccurred": "2025-02-11T08:16:16Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-02-11T08:16:16Z",
  "eventType": "InformationRequired",
  "event": {
    "resourceOrder": {
      "@type": "ResourceOrder",
      "id": "5130d5131a23",
      "href": "http://servername/resourceOrder/5130d5131a23",
      "category": "Premium",
      "description": "7000m of Optical Cable",
      "expectedCompletionDate": "2020-05-31T20:50:27.275Z",
      "name": "7km Optical Cable for Acme Inc",
      "priority": 0,
```

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```
      "requestedCompletionDate": "2020-03-31T19:50:43.214Z",
      "requestedStartDate": "2020-03-31T19:50:43.214Z",
      "startDate": "2020-03-31T19:50:43.214Z"
    },
    "informationRequired": {
      "id": "info-req-12345",
      "href": "http://servername/informationRequired/info-req-12345",
      "@type": "InformationRequiredEvent",
      "informationRequiredPath": [
        {
          "op": "add",
          "path": "/orderItem/0/resource/specification",
          "value": "Resource specification is required"
        }
      ]
    }
  }
}
```


State Change Event

Notification sent when a resource order transitions to a new state


```
Content-Type: application/json
{
  "@type": "ResourceOrderStateChangeEvent",
  "@schemaLocation": "/schema-registry/schema/resourceOrderStateChangeEvent",
  "href": "/tmf-api/v5/events/stateChange/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Resource Order State Change",
  "description": "This event captures changes made to the state of a resource order.",
  "priority": "medium",
  "timeOccurred": "2025-01-22T06:41:27Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
```


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```
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "char123",
      "name": "OrderPriorityChange",
      "valueType": "string",
      "value": "High",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "rel456",
          "relationshipType": "associated"
        }
      ]
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-01-22T06:41:27Z",
  "eventType": "StateChange",
  "event": {
    "resourceOrder": {
      "@type": "ResourceOrder",
      "id": "12345",
      "href": "/tmf-api/v5/resourceOrder/12345",
      "state": "acknowledged"
    }
  }
}
```


CancelResourceOrder

Create Event

Notification sent when a new cancel resource order request is created


```
Content-Type: application/json
{
  "@type": "CancelResourceOrderCreateEvent",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrderCreateEvent",
  "href": "/tmf-api/events/cancelResourceOrder/event12345",
  "id": "event12345",
  "correlationId": "corr98765",
  "domain": "ResourceOrderManagement",
  "title": "Cancel Resource Order Event",
  "description": "This event indicates the cancellation of a resource order.",
  "priority": "high",
```


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```
  "timeOccurred": "2025-01-22T06:41:26Z",
  "source": {
    "@type": "Source",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/systems/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/systems/reportingSystem/report123",
    "id": "report123",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/parties/party123",
        "id": "party123",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Operator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/parties/party456",
        "id": "party456",
        "name": "Jane Smith",
        "@referredType": "Organization"
      }
    }
  ],
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "char001",
      "name": "CancellationReason",
      "valueType": "string",
      "value": "CustomerRequest",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "rel001",
          "relationshipType": "dependent"
        }
      ]
    }
```


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```
  ],
  "eventId": "event001",
  "eventTime": "2025-01-22T06:41:26Z",
  "eventType": "CancelResourceOrder",
  "event": {
    "cancelResourceOrder": {
      "@type": "CancelResourceOrder",
      "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
      "requestedCancellationDate": "2025-01-22T06:41:26Z",
      "effectiveCancellationDate": "2025-01-22T06:41:26Z",
      "cancellationReason": "CustomerRequest",
      "state": "acknowledged",
      "resourceOrder": {
        "@type": "ResourceOrderRef",
        "@schemaLocation": "/schema-registry/schema/ResourceOrder",
        "href": "/tmf-api/resourceOrder/RO123",
        "id": "RO123",
        "name": "ResourceOrder123",
        "@referredType": "ResourceOrder"
      }
    }
  }
}
```

Information Required Event

Notification sent when additional information is needed to process a cancellation request

```
Content-Type: application/json
{
  "@type": "CancelResourceOrderInformationRequiredEvent",
  "@schemaLocation": "/schema-
registry/schema/cancelResourceOrderInformationRequiredEvent",
  "href": "/tmf-api/v5/events/informationRequired/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Cancel Resource Order Information Required",
  "description": "This event captures information required for a cancel resource
order.",
  "priority": "medium",
  "timeOccurred": "2025-02-11T08:40:56Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
```

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```
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-02-11T08:40:56Z",
  "eventType": "InformationRequired",
  "event": {
    "cancelResourceOrder": {
      "@type": "CancelResourceOrder",
      "id": "cancel-order-5130d51",
      "href": "http://servername/cancelResourceOrder/cancel-order-5130d51"
    },
    "informationRequired": {
      "id": "info-req-cancel-12345",
      "href": "http://servername/informationRequired/info-req-cancel-12345",
      "@type": "InformationRequiredEvent",
      "informationRequiredPath": [
        {
          "op": "add",
          "path": "/cancellationReason",
          "value": "Cancellation reason is required"
        }
      ]
    }
  }
}
```


State Change Event

Notification sent when a cancel resource order request transitions to a new state


```
Content-Type: application/json
{
  "@type": "CancelResourceOrderStateChangeEvent",
  "@baseType": "Event",
  "@schemaLocation": "/schema-registry/schema/cancelResourceOrderStateChangeEvent",
  "href": "/tmf-api/v5/events/stateChange/12345",
  "id": "12345",
  "correlationId": "corr67890",
  "domain": "OrderManagement",
  "title": "Cancel Resource Order State Change",
  "description": "This event captures changes made to the state of a cancel resource
order.",
  "priority": "medium",
```


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```
  "timeOccurred": "2025-01-22T06:41:27Z",
  "source": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/System",
    "href": "/tmf-api/v5/systems/sourceSystem/source123",
    "id": "source123",
    "name": "SourceSystem"
  },
  "reportingSystem": {
    "@type": "System",
    "@schemaLocation": "/schema-registry/schema/ReportingSystem",
    "href": "/tmf-api/v5/systems/reportingSystem/report567",
    "id": "report567",
    "name": "ReportingSystem"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/customer789",
        "id": "customer789",
        "name": "John Doe",
        "@referredType": "Individual"
      }
    }
  ],
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "char123",
      "name": "OrderStateChange",
      "valueType": "string",
      "value": "acknowledged",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "rel456",
          "relationshipType": "associated"
        }
      ]
    }
  ],
  "eventId": "event56789",
  "eventTime": "2025-01-22T06:41:27Z",
  "eventType": "StateChange",
  "event": {
    "cancelResourceOrder": {
      "@type": "CancelResourceOrder",
      "id": "cancel-order-12345",
      "href": "http://servername/cancelResourceOrder/cancel-order-12345",
      "state": "acknowledged"
    }
  }
```


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```
}
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ResourceOrder

List or find ResourceOrder objects

`GET /resourceOrder?fields=…`

Description

This operation list ResourceOrder entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of resource orders with date range filtering, sorting, and pagination

Request

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```
GET /resourceOrder?offset=0&limit=50&before=2025-01-20T00:00:00Z&after=2025-01-
01T00:00:00Z&sort=creationDate:desc
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ResourceOrder",
    "href": "https://host:port/tmf-api/resourceOrdering/v5/resourceOrder/PWBSW",
    "id": "PWBSW",
    "creationDate": "2025-01-22T06:41:23Z",
    "cancellationDate": "2025-01-22T06:41:23Z",
    "cancellationReason": "Customer requested cancellation due to project
termination.",
    "externalId": [
      {
        "@type": "ExternalIdentifier",
        "id": "EXT00001",
        "owner": "ExternalSystem",
        "externalIdentifierType": "OrderReference"
      }
    ],
    "name": "New Network Deployment Order",
    "description": "This order pertains to the deployment of a new network
infrastructure.",
    "priority": 1,
    "orderType": "Deployment",
    "category": "Infrastructure",
    "requestedStartDate": "2025-01-22T06:41:23Z",
    "requestedCompletionDate": "2025-02-15T06:41:23Z",
    "startDate": "2025-01-23T06:41:23Z",
    "completionDate": "2025-02-10T06:41:23Z",
    "expectedCompletionDate": "2025-02-12T06:41:23Z",
    "note": [
      {
        "@type": "Note",
        "id": "NOTE001",
        "author": "John Doe",
        "date": "2025-01-22T06:41:23Z",
        "text": "Order created and acknowledged by the operations team."
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation": "/schema-registry/schema/RelatedParty",
        "role": "Service Provider",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "/schema-registry/schema/Party",
          "href": "/tmf-api/v5/party/party123",
          "id": "party123",
          "name": "Service Provider Organization",
          "@referredType": "Organization"
```

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```
        }
      }
    ],
    "state": "acknowledged",
    "orderItem": [
      {
        "@type": "ResourceOrderItem",
        "orderItemRelationship": [
          {
            "@type": "ResourceOrderItemRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
            "relationshipType": "dependency",
            "orderItem": {
              "@type": "ResourceOrderItemRef",
              "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
              "itemId": "ITEM56789",
              "resourceOrderId": "ORDER12345",
              "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
              "@referredType": "ResourceOrderItem"
            }
          }
        ],
        "quantity": 5,
        "appointment": {
          "@type": "AppointmentRef",
          "@schemaLocation": "/schema-registry/schema/AppointmentRef",
          "href": "/tmf-api/v5/appointment/67890",
          "id": "67890",
          "name": "Site Installation Appointment",
          "@referredType": "Appointment",
          "description": "Appointment for the installation of network infrastructure at
the customer site."
        },
        "action": "add",
        "id": "12345",
        "resource": {
          "@type": "PhysicalResource",
          "@baseType": "Resource",
          "@schemaLocation": "/schema-registry/schema/Resource",
          "href": "/tmf-api/v5/resource/98765",
          "id": "98765",
          "category": "Network Element",
          "description": "Fiber Optic Cable connecting two locations for high-speed
data transmission.",
          "name": "Fiber Optic Cable - Main Line",
          "endOperatingDate": "2025-01-22T06:41:23Z",
          "administrativeState": "locked",
          "operationalState": "enabled",
          "usageState": "idle",
          "validFor": {
            "startDateTime": "2025-01-22T06:41:23Z",
            "endDateTime": "2025-02-25T06:41:23Z"
          },
          "note": [
            {
              "@type": "Note",
              "id": "12345",
              "author": "John Doe",
              "date": "2025-01-22T06:41:23Z",
              "text": "This resource order was created to address a high-priority
```


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```
customer request for infrastructure upgrade."
            }
          ],
          "resourceOrderItem": [
            {
              "@type": "RelatedResourceOrderItem",
              "@schemaLocation": "/schema-registry/schema/ResourceOrderItem",
              "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
              "resourceOrderId": "RO12345",
              "itemAction": "add",
              "itemId": "ITEM67890",
              "role": "primary",
              "@referredType": "ResourceOrderItem"
            }
          ],
          "place": [
            {
              "@type": "RelatedPlaceRef",
              "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
              "role": "installation",
              "place": {
                "@type": "PlaceRef",
                "@schemaLocation": "/schema-registry/schema/PlaceRef",
                "href": "/tmf-api/v5/place/98765",
                "id": "98765",
                "name": "Customer Site - Location 1",
                "@referredType": "GeographicalLocation"
              }
            }
          ],
          "relatedParty": [
            {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
              "role": "Service Provider",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "@referredType": "Party",
                "@schemaLocation": "/schema-registry/schema/PartyRef",
                "href": "/tmf-api/v5/party/98765",
                "id": "98765",
                "name": "ABC Telecom Services"
              }
            }
          ],
          "supportingResource": [
            {
              "@type": "PhysicalResource",
              "@baseType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/54321",
              "id": "54321",
              "category": "Networking Equipment",
              "description": "Router used to manage network traffic between different
segments of the network.",
              "name": "Main Router - Data Center",
              "endOperatingDate": "2025-01-22T06:41:23Z",
              "administrativeState": "locked",
              "operationalState": "enabled",
```


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```
              "usageState": "idle",
              "validFor": {
                "startDateTime": "2025-01-22T06:41:23Z",
                "endDateTime": "2025-01-22T06:41:23Z"
              },
              "note": [
                {
                  "@type": "Note",
                  "id": "12345",
                  "author": "John Doe",
                  "date": "2025-01-22T06:41:23Z",
                  "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
                }
              ],
              "resourceOrderItem": [
                {
                  "@type": "RelatedResourceOrderItem",
                  "@schemaLocation": "/schema-
registry/schema/RelatedResourceOrderItem",
                  "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                  "resourceOrderId": "RO12345",
                  "itemAction": "add",
                  "itemId": "ITEM67890",
                  "role": "primary",
                  "@referredType": "ResourceOrderItem"
                }
              ],
              "place": [
                {
                  "@type": "RelatedPlaceRef",
                  "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                  "role": "installation",
                  "place": {
                    "@type": "PlaceRef",
                    "@schemaLocation": "/schema-registry/schema/PlaceRef",
                    "href": "/tmf-api/v5/place/98765",
                    "id": "98765",
                    "name": "Customer Site - Location 1",
                    "@referredType": "GeographicalLocation"
                  }
                }
              ],
              "relatedParty": [
                {
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                  "role": "Service Provider",
                  "partyOrPartyRole": {
                    "@type": "PartyRef",
                    "@referredType": "Party",
                    "@schemaLocation": "/schema-registry/schema/PartyRef",
                    "href": "/tmf-api/v5/party/98765",
                    "id": "98765",
                    "name": "ABC Telecom Services"
                  }
                }
              ],
              "resourceRelationship": [
```


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```
                {
                  "@type": "ResourceRelationship",
                  "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                  "resourceRelationshipCharacteristic": [
                    {
                      "@type": "IntegerCharacteristic",
                      "@baseType": "Characteristic",
                      "@schemaLocation": "/schema-registry/schema/Characteristic",
                      "id": "12345",
                      "name": "Performance Level",
                      "valueType": "Integer",
                      "value": 5,
                      "characteristicRelationship": [
                        {
                          "@type": "CharacteristicRelationship",
                          "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                          "id": "123",
                          "relationshipType": "isRelatedTo"
                        }
                      ]
                    }
                  ],
                  "resource": {
                    "@type": "ResourceRef",
                    "id": "56789",
                    "@referredType": "Resource",
                    "@schemaLocation": "/schema-registry/schema/Resource",
                    "href": "/tmf-api/v5/resource/56789",
                    "name": "Network Equipment"
                  },
                  "relationshipType": "dependsOn"
                }
              ],
              "resourceCharacteristic": [
                {
                  "@type": "IntegerCharacteristic",
                  "@baseType": "Characteristic",
                  "@schemaLocation": "/schema-registry/schema/Characteristic",
                  "id": "12345",
                  "name": "Processing Speed",
                  "valueType": "Integer",
                  "value": 1000,
                  "characteristicRelationship": [
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "12345",
                      "relationshipType": "affects"
                    }
                  ]
                }
              ],
              "attachment": [
                {
                  "@type": "AttachmentRef",
                  "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                  "href": "/tmf-api/v5/attachment/12345",
                  "id": "12345",
```


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```
                  "name": "Network Upgrade Plan",
                  "@referredType": "Attachment",
                  "description": "Detailed documentation for the upcoming network
upgrade project.",
                  "url": "http://example.com/documents/network-upgrade"
                }
              ],
              "resourceSpecification": {
                "@type": "ResourceSpecificationRef",
                "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
                "href": "/tmf-api/v5/specification/12345",
                "id": "12345",
                "name": "High-Speed Router",
                "@referredType": "ResourceSpecification",
                "version": "1.0"
              },
              "startOperatingDate": "2025-01-22T06:41:23Z",
              "resourceVersion": "2.0",
              "activationFeature": [
                {
                  "@type": "Feature",
                  "@schemaLocation": "/schema-registry/schema/Feature",
                  "isBundle": true,
                  "featureRelationship": [
                    {
                      "@type": "FeatureRelationship",
                      "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                      "href": "/tmf-api/v5/feature/12345",
                      "id": "12345",
                      "name": "Advanced Security",
                      "relationshipType": "excluded",
                      "validFor": {
                        "startDateTime": "2025-01-22T06:41:23Z",
                        "endDateTime": "2025-01-22T06:41:23Z"
                      }
                    }
                  ],
                  "featureCharacteristic": [
                    {
                      "@type": "IntegerCharacteristic",
                      "@baseType": "Characteristic",
                      "@schemaLocation": "/schema-registry/schema/Characteristic",
                      "id": "12345",
                      "name": "Firewall Strength",
                      "valueType": "Integer",
                      "value": 10,
                      "characteristicRelationship": [
                        {
                          "@type": "CharacteristicRelationship",
                          "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                          "id": "12345",
                          "relationshipType": "enhances"
                        }
                      ]
                    }
                  ],
                  "policyConstraint": [
                    {
                      "@type": "PolicyRef",
```


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```
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "/tmf-api/v5/policy/12345",
                    "id": "12345",
                    "name": "Service Level Agreement",
                    "@referredType": "Policy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": true,
                "id": "12345",
                "name": "Firewall Protection"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "/tmf-api/v5/intent/12345",
              "id": "12345",
              "name": "Security Enhancement",
              "@referredType": "Intent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 10000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
```


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```
                "@schemaLocation": "/schema-registry/schema/Resource",
                "href": "/tmf-api/v5/resource/87654",
                "name": "Server 01"
              },
              "relationshipType": "dependent"
            }
          ],
          "resourceCharacteristic": [
            {
              "@type": "IntegerCharacteristic",
              "@baseType": "Characteristic",
              "@schemaLocation": "/schema-registry/schema/Characteristic",
              "id": "CH123456",
              "name": "Storage Capacity",
              "valueType": "integer",
              "value": 5000,
              "characteristicRelationship": [
                {
                  "@type": "CharacteristicRelationship",
                  "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                  "id": "CHREL7890",
                  "relationshipType": "dependent"
                }
              ]
            }
          ],
          "attachment": [
            {
              "@type": "AttachmentRef",
              "@schemaLocation": "/schema-registry/schema/Attachment",
              "href": "/tmf-api/v5/attachment/ATT98765",
              "id": "ATT98765",
              "name": "Product Specification",
              "@referredType": "Attachment",
              "description": "Detailed product specification document.",
              "url": "http://example.com/downloads/product-spec"
            }
          ],
          "resourceSpecification": {
            "@type": "ResourceSpecificationRef",
            "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
            "href": "/tmf-api/v5/specification/SPEC54321",
            "id": "SPEC54321",
            "name": "Enterprise Storage",
            "@referredType": "ResourceSpecification",
            "version": "2.1"
          },
          "startOperatingDate": "2025-01-22T06:41:23Z",
          "resourceVersion": "3.0",
          "activationFeature": [
            {
              "@type": "Feature",
              "@schemaLocation": "/schema-registry/schema/Feature",
              "isBundle": false,
              "featureRelationship": [
                {
                  "@type": "FeatureRelationship",
                  "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                  "href": "/tmf-api/v5/feature/12345",
```


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```
                  "id": "12345",
                  "name": "Network Upgrade",
                  "relationshipType": "excluded",
                  "validFor": {
                    "startDateTime": "2025-01-22T06:41:23Z",
                    "endDateTime": "2025-01-22T06:41:23Z"
                  }
                }
              ],
              "featureCharacteristic": [
                {
                  "@type": "IntegerCharacteristic",
                  "@baseType": "Characteristic",
                  "@schemaLocation": "/schema-registry/schema/Characteristic",
                  "id": "45678",
                  "name": "Speed",
                  "valueType": "Integer",
                  "value": 1000,
                  "characteristicRelationship": [
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "34567",
                      "relationshipType": "associated"
                    }
                  ]
                }
              ],
              "policyConstraint": [
                {
                  "@type": "PolicyRef",
                  "@schemaLocation": "/schema-registry/schema/PolicyRef",
                  "href": "/tmf-api/v5/policy/12345",
                  "id": "12345",
                  "name": "Data Retention Policy",
                  "@referredType": "Policy",
                  "version": "1.0"
                }
              ],
              "isEnabled": false,
              "id": "23456",
              "name": "Service Bundle"
            }
          ],
          "intent": {
            "@type": "IntentRef",
            "@schemaLocation": "/schema-registry/schema/IntentRef",
            "href": "/tmf-api/v5/intent/INT1001",
            "id": "INT1001",
            "name": "Upgrade Service",
            "@referredType": "Intent"
          },
          "externalIdentifier": [
            {
              "@type": "ExternalIdentifier",
              "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
              "owner": "XYZ Corp",
              "externalIdentifierType": "ProjectID",
              "id": "EXT45678"
```


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```
            }
          ]
        },
        "state": "acknowledged"
      }
    ]
  }
]
```

Retrieves a ResourceOrder by ID

`GET /resourceOrder/{id}?fields=…`

Description

This operation retrieves a ResourceOrder entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a specific resource order using its unique identifier

Request

```
GET /resourceOrder/12345
Content-Type: application/json
```

Response

```
200
{
  "@type": "ResourceOrder",
  "href": "https://host:port/tmf-api/resourceOrdering/v5/resourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/ResourceOrder",
  "creationDate": "2025-01-22T06:41:23Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "id": "EXT00001",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "This order pertains to the deployment of a new network
infrastructure.",
  "priority": 1,
  "orderType": "Deployment",
  "category": "Infrastructure",
```

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```
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "expectedCompletionDate": "2025-02-12T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T06:41:23Z",
      "text": "Order created and acknowledged by the operations team."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 1,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "Appointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
```


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```
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary",
            "@referredType": "ResourceOrderItem"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "GeographicalLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
```


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```
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "Organization"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary",
                "@referredType": "ResourceOrderItem"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
```


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```
                  "name": "Customer Site - Location 1",
                  "@referredType": "GeographicalLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "Party"
                }
              }
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
                  "@type": "ResourceRef",
                  "@referredType": "Resource",
                  "id": "56789",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
```


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```
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "TechnicalDocumentation",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
```


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```
                {
                  "@type": "IntegerCharacteristic",
                  "@baseType": "Characteristic",
                  "@schemaLocation": "/schema-registry/schema/Characteristic",
                  "id": "12345",
                  "name": "Firewall Strength",
                  "valueType": "Integer",
                  "value": 5,
                  "characteristicRelationship": [
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "12345",
                      "relationshipType": "enhances"
                    }
                  ]
                }
              ],
              "policyConstraint": [
                {
                  "@type": "PolicyRef",
                  "@schemaLocation": "/schema-registry/schema/PolicyRef",
                  "href": "/tmf-api/v5/policy/12345",
                  "id": "12345",
                  "name": "Service Level Agreement",
                  "@referredType": "ServicePolicy",
                  "version": "1.0"
                }
              ],
              "isEnabled": true,
              "id": "12345",
              "name": "Firewall Protection"
            }
          ],
          "intent": {
            "@type": "IntentRef",
            "@schemaLocation": "/schema-registry/schema/IntentRef",
            "href": "/tmf-api/v5/intent/12345",
            "id": "12345",
            "name": "Security Enhancement",
            "@referredType": "Intent"
          },
          "externalIdentifier": [
            {
              "@type": "ExternalIdentifier",
              "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
              "owner": "SecurityInc",
              "externalIdentifierType": "OrderReference",
              "id": "12345"
            }
          ]
        }
      ],
      "resourceRelationship": [
        {
          "@type": "ResourceRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
          "resourceRelationshipCharacteristic": [
            {
```


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```
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 5000,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Attachment",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
```


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```
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "DataPolicy",
                "version": "1.0"
```


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```
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "ServiceUpgrade"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}
```


Creates a ResourceOrder

`POST /resourceOrder?fields=…`

Description

This operation creates a ResourceOrder entity.

Mandatory Attributes

Mandatory Attributes	Rule
note.id	
note.text	
orderItem	
orderItem.action	
orderItem.appointment.id	
orderItem.id	

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Mandatory Attributes	Rule
orderItem.orderItemRelationship.orderItem	
orderItem.orderItemRelationship.orderItem.itemId	
relatedParty.partyOrPartyRole.@referredType	
relatedParty.partyOrPartyRole.id	
relatedParty.role	
@type	M

Usage samples

Create a new resource order with multiple order items and related parties

Request

```
POST /resourceOrder
Content-Type: application/json
{
  "@type": "ResourceOrder",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
```

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```
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary",
            "@referredType": "ResourceOrderItem"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "GeographicalLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "Party"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
```


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```
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary",
                "@referredType": "ResourceOrderItem"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "GeographicalLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "Party"
                }
              }
```


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```
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "resource": {
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ]
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
```


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```
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "Attachment",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "value": 10,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "12345",
                        "relationshipType": "enhances"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
```


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```
                {
                  "@type": "PolicyRef",
                  "@schemaLocation": "/schema-registry/schema/PolicyRef",
                  "href": "/tmf-api/v5/policy/12345",
                  "id": "12345",
                  "name": "Service Level Agreement",
                  "@referredType": "Policy",
                  "version": "1.0"
                }
              ],
              "isEnabled": true,
              "id": "12345",
              "name": "Firewall Protection"
            }
          ],
          "intent": {
            "@type": "IntentRef",
            "@schemaLocation": "/schema-registry/schema/IntentRef",
            "href": "/tmf-api/v5/intent/12345",
            "id": "12345",
            "name": "Security Enhancement",
            "@referredType": "Intent"
          },
          "externalIdentifier": [
            {
              "@type": "ExternalIdentifier",
              "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
              "owner": "SecurityInc",
              "externalIdentifierType": "OrderReference",
              "id": "12345"
            }
          ]
        }
      ],
      "resourceRelationship": [
        {
          "@type": "ResourceRelationship",
          "resource": {
            "@type": "ResourceRef",
            "id": "87654",
            "@referredType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/87654",
            "name": "Server 01"
          },
          "relationshipType": "dependent",
          "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
          "resourceRelationshipCharacteristic": [
            {
              "@type": "IntegerCharacteristic",
              "@baseType": "Characteristic",
              "@schemaLocation": "/schema-registry/schema/Characteristic",
              "id": "12345",
              "name": "Bandwidth",
              "valueType": "Integer",
              "value": 10000,
              "characteristicRelationship": [
                {
                  "@type": "CharacteristicRelationship",
                  "@schemaLocation": "/schema-
```


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```
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ]
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 5000,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Attachment",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
```

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```
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "Policy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "Intent"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
```


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```
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "OrderItem"
          },
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency"
        }
      ],
      "quantity": 150,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "Appointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      }
    }
  ],
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "id": "XTVMA",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    }
  ],
  "name": "Network Resource Order",
  "description": "This order is to provision network resources for the client.",
  "priority": 5,
  "orderType": "Provisioning",
  "category": "Network",
  "requestedStartDate": "2025-01-22T08:00:00Z",
  "requestedCompletionDate": "2025-01-25T18:00:00Z",
  "startDate": "2025-01-22T08:00:00Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T08:00:00Z",
      "text": "This order requires expedited processing due to client SLA."
    }
  ],
  "relatedParty": [
```


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```
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "@schemaLocation": "/schema-registry/schema/related-party",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ]
}
```

Response

```
201
{
  "@type": "ResourceOrder",
  "href": "/tmf-api/v5/resourceOrder/12345",
  "id": "12345",
  "creationDate": "2025-01-22T06:41:23Z",
  "cancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested cancellation due to project termination.",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "id": "EXT00001",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    },
    {
      "@type": "ExternalIdentifier",
      "id": "EXT00002",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "This order pertains to the deployment of a new network
infrastructure.",
  "priority": 1,
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "startDate": "2025-01-23T06:41:23Z",
  "completionDate": "2025-02-10T06:41:23Z",
  "expectedCompletionDate": "2025-02-12T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T06:41:23Z",
```

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```
      "text": "Order created and acknowledged by the operations team."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 10,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "Appointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
```


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```
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary",
            "@referredType": "ResourceOrderItem"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "GeographicalLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "Party"
            }
          }
        ],
```


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```
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary",
                "@referredType": "ResourceOrderItem"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "GeographicalLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
```


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```
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "Party"
                }
              }
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCcharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 10,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
```

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```
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "Attachment",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "value": 8,
                    "characteristicRelationship": [
```


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```
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "12345",
                      "relationshipType": "enhances"
                    }
                  ]
                }
              ],
              "policyConstraint": [
                {
                  "@type": "PolicyRef",
                  "@schemaLocation": "/schema-registry/schema/PolicyRef",
                  "href": "/tmf-api/v5/policy/12345",
                  "id": "12345",
                  "name": "Service Level Agreement",
                  "@referredType": "Policy",
                  "version": "1.0"
                }
              ],
              "isEnabled": true,
              "id": "12345",
              "name": "Firewall Protection"
            }
          ],
          "intent": {
            "@type": "IntentRef",
            "@schemaLocation": "/schema-registry/schema/IntentRef",
            "href": "/tmf-api/v5/intent/12345",
            "id": "12345",
            "name": "Security Enhancement",
            "@referredType": "Intent"
          },
          "externalIdentifier": [
            {
              "@type": "ExternalIdentifier",
              "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
              "owner": "SecurityInc",
              "externalIdentifierType": "OrderReference",
              "id": "12345"
            }
          ]
        }
      ],
      "resourceRelationship": [
        {
          "@type": "ResourceRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
          "resourceRelationshipCharacteristic": [
            {
              "@type": "IntegerCharacteristic",
              "@baseType": "Characteristic",
              "@schemaLocation": "/schema-registry/schema/Characteristic",
              "id": "12345",
              "name": "Bandwidth",
              "valueType": "Integer",
              "value": 1000,
              "characteristicRelationship": [
                {
```


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```
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 500,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Attachment",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
```


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```
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "@referredType": "NetworkFeature",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 100,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "Policy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
```


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```
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "Intent"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}
```


Updates partially a ResourceOrder

`PATCH /resourceOrder/{id}?fields=…`

Description

This operation allows partial updates of a ResourceOrder entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
completionDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

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Patchable Attributes	Rule
cancellationDate	
cancellationReason	
category	
creationDate	
description	
expectedCompletionDate	
externalId	
name	
note	
orderItem	
orderType	
priority	
relatedParty	
requestedCompletionDate	
requestedStartDate	
startDate	
state	

Usage samples

Partially update an existing resource order using standard JSON merge patch

Request

```
PATCH /resourceOrder/EXNHG
Content-Type: application/json
{
  "@type": "ResourceOrder",
  "description": "Updated description: This order pertains to the deployment of a new
network infrastructure with enhanced security features.",
  "priority": 2,
  "expectedCompletionDate": "2025-02-20T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE002",
      "author": "Jane Smith",
      "date": "2025-01-25T10:30:00Z",
```

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```
      "text": "Updated order priority and completion date due to additional security
requirements."
    }
  ]
}
```

Response

```
200
{
  "@type": "ResourceOrder",
  "href": "https://host:port/tmf-api/resourceOrdering/v5/resourceOrder/EXNHG",
  "id": "EXNHG",
  "@schemaLocation": "/schema-registry/schema/ResourceOrder",
  "creationDate": "2025-01-22T06:41:23Z",
  "cancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested cancellation due to project termination.",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "owner": "ExternalSystem",
      "externalIdentifierType": "ContractReference",
      "id": "EXT123456"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "Updated description: This order pertains to the deployment of a new
network infrastructure with enhanced security features.",
  "priority": 2,
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "expectedCompletionDate": "2025-02-20T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE002",
      "author": "Jane Smith",
      "date": "2025-01-25T10:30:00Z",
      "text": "Updated order priority and completion date due to additional security
requirements."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
```

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```
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 1,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "InstallationAppointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
```


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```
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "@referredType": "ResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "SiteLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "TelecommunicationsProvider"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
```


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```
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "@referredType": "ResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "SiteLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "TelecommunicationsProvider"
                }
              }
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
```


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```
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "TechnicalDocumentation",
                "description": "Detailed documentation for the upcoming network upgrade
```


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```
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "value": 10,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "12345",
                        "relationshipType": "enhances"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
                  {
                    "@type": "PolicyRef",
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "/tmf-api/v5/policy/12345",
                    "id": "12345",
```


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```
                    "name": "Service Level Agreement",
                    "@referredType": "ServicePolicy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": true,
                "id": "12345",
                "name": "Firewall Protection"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "/tmf-api/v5/intent/12345",
              "id": "12345",
              "name": "Security Enhancement",
              "@referredType": "ServiceIntent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
```


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```
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 1000,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Document",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "relationshipType": "excluded",
```


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```
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "DataPolicy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "ServiceUpgrade"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
```


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```
      "state": "acknowledged"
    }
  ]
}
```

Update a resource order using RFC 6902 JSON Patch operations (add, remove, replace)

Request

```
PATCH /resourceOrder/12345
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/priority",
    "value": 2
  },
  {
    "op": "replace",
    "path": "/description",
    "value": "Updated description: This order pertains to the deployment of a new
network infrastructure with enhanced security features."
  },
  {
    "op": "add",
    "path": "/note/-",
    "value": {
      "@type": "Note",
      "id": "NOTE002",
      "author": "Jane Smith",
      "date": "2025-01-25T10:30:00Z",
      "text": "Updated order priority and completion date due to additional security
requirements."
    }
  },
  {
    "op": "replace",
    "path": "/expectedCompletionDate",
    "value": "2025-02-20T06:41:23Z"
  }
]
```

Response

```
200
{
  "@type": "ResourceOrder",
  "href": "https://host:port/tmf-api/resourceOrdering/v5/resourceOrder/YYRBC",
  "id": "YYRBC",
  "creationDate": "2025-01-22T06:41:23Z",
  "cancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested cancellation due to project termination.",
  "externalId": [
    {
```

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```
      "@type": "ExternalIdentifier",
      "id": "EXT00001",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "This order pertains to the deployment of a new network
infrastructure.",
  "priority": 1,
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "startDate": "2025-01-23T06:41:23Z",
  "completionDate": "2025-02-10T06:41:23Z",
  "expectedCompletionDate": "2025-02-12T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T06:41:23Z",
      "text": "Order created and acknowledged by the operations team."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
```


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```
      "quantity": 10,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "Appointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary",
            "@referredType": "ResourceOrderItem"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
```


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```
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "GeographicalLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "Party"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
```


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```
                "role": "primary",
                "@referredType": "ResourceOrderItem"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "GeographicalLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "Party"
                }
              }
            ],
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
```


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```
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 2000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "Attachment",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
```


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```
                {
                  "@type": "FeatureRelationship",
                  "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                  "href": "/tmf-api/v5/feature/12345",
                  "id": "12345",
                  "name": "Advanced Security",
                  "relationshipType": "excluded",
                  "validFor": {
                    "startDateTime": "2025-01-22T06:41:23Z",
                    "endDateTime": "2025-01-22T06:41:23Z"
                  }
                }
              ],
              "featureCharacteristic": [
                {
                  "@type": "IntegerCharacteristic",
                  "@baseType": "Characteristic",
                  "@schemaLocation": "/schema-registry/schema/Characteristic",
                  "id": "12345",
                  "name": "Firewall Strength",
                  "valueType": "Integer",
                  "value": 10,
                  "characteristicRelationship": [
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "12345",
                      "relationshipType": "enhances"
                    }
                  ]
                }
              ],
              "policyConstraint": [
                {
                  "@type": "PolicyRef",
                  "@schemaLocation": "/schema-registry/schema/PolicyRef",
                  "href": "/tmf-api/v5/policy/12345",
                  "id": "12345",
                  "name": "Service Level Agreement",
                  "@referredType": "Policy",
                  "version": "1.0"
                }
              ],
              "isEnabled": true,
              "id": "12345",
              "name": "Firewall Protection"
            }
          ],
          "intent": {
            "@type": "IntentRef",
            "@schemaLocation": "/schema-registry/schema/IntentRef",
            "href": "/tmf-api/v5/intent/12345",
            "id": "12345",
            "name": "Security Enhancement",
            "@referredType": "Intent"
          },
          "externalIdentifier": [
            {
              "@type": "ExternalIdentifier",
```


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```
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 10000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 5000,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
```


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```
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Attachment",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
```


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```
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "Policy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "Intent"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}
```

Update a resource order using JSON Patch with query capabilities for complex updates

Request

```
PATCH /resourceOrder/67890
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/note?(@.id=='NOTE001')/text",
    "value": "Order updated - priority escalated due to customer requirements."
  },
  {
```

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```
    "op": "replace",
    "path": "/orderItem/-/resource/resourceCharacteristic/-",
    "value": {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/schema/Characteristic",
      "id": "CH123456",
      "name": "Storage Capacity",
      "valueType": "integer",
      "value": 2048,
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "@schemaLocation": "/schema-registry/schema/CharacteristicRelationship",
          "id": "CHREL7890",
          "relationshipType": "dependent"
        }
      ]
    }
  },
  {
    "op": "replace",
    "path": "/relatedParty/-",
    "value": {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/relatedParties/67890",
        "id": "67890",
        "name": "ABC Telecom - Premium Services",
        "@referredType": "Organization"
      }
    }
  }
]
```

Response

```
200
{
  "@type": "ResourceOrder",
  "href": "/tmf-api/v5/resourceOrder/12345",
  "id": "12345",
  "creationDate": "2025-01-22T06:41:23Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "id": "EXT00001",
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    },
    {
      "@type": "ExternalIdentifier",
      "id": "EXT00002",
```

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```
      "owner": "ExternalSystem",
      "externalIdentifierType": "OrderReference"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "This order pertains to the deployment of a new network
infrastructure.",
  "priority": 1,
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "startDate": "2025-01-23T06:41:23Z",
  "completionDate": "2025-02-10T06:41:23Z",
  "expectedCompletionDate": "2025-02-12T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE001",
      "author": "John Doe",
      "date": "2025-01-22T06:41:23Z",
      "text": "Order created and acknowledged by the operations team."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 1,
      "appointment": {
```


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```
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "Appointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary",
            "@referredType": "ResourceOrderItem"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
```


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```
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "GeographicalLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "Party"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary",
                "@referredType": "ResourceOrderItem"
```


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```
            }
          ],
          "place": [
            {
              "@type": "RelatedPlaceRef",
              "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
              "role": "installation",
              "place": {
                "@type": "PlaceRef",
                "@schemaLocation": "/schema-registry/schema/PlaceRef",
                "href": "/tmf-api/v5/place/98765",
                "id": "98765",
                "name": "Customer Site - Location 1",
                "@referredType": "GeographicalLocation"
              }
            }
          ],
          "relatedParty": [
            {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
              "role": "Service Provider",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "@schemaLocation": "/schema-registry/schema/PartyRef",
                "href": "/tmf-api/v5/party/98765",
                "id": "98765",
                "name": "ABC Telecom Services",
                "@referredType": "Party"
              }
            }
          ],
          "resourceRelationship": [
            {
              "@type": "ResourceRelationship",
              "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
              "resourceRelationshipCharacteristic": [
                {
                  "@type": "IntegerCharacteristic",
                  "@baseType": "Characteristic",
                  "@schemaLocation": "/schema-registry/schema/Characteristic",
                  "id": "12345",
                  "name": "Performance Level",
                  "valueType": "Integer",
                  "value": 5,
                  "characteristicRelationship": [
                    {
                      "@type": "CharacteristicRelationship",
                      "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                      "id": "123",
                      "relationshipType": "isRelatedTo"
                    }
                  ]
                }
              ],
              "resource": {
                "@type": "ResourceRef",
                "id": "56789",
```


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```
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 10,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "Attachment",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
```


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```
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "value": 8,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "12345",
                        "relationshipType": "enhances"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
                  {
                    "@type": "PolicyRef",
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "/tmf-api/v5/policy/12345",
                    "id": "12345",
                    "name": "Service Level Agreement",
                    "@referredType": "Policy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": true,
                "id": "12345",
                "name": "Firewall Protection"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "/tmf-api/v5/intent/12345",
              "id": "12345",
              "name": "Security Enhancement",
              "@referredType": "Intent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
```


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```
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 500,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
```


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```
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Attachment",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "@referredType": "NetworkFeature",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 100,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
```


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```
                }
              ]
            }
          ],
          "policyConstraint": [
            {
              "@type": "PolicyRef",
              "@schemaLocation": "/schema-registry/schema/PolicyRef",
              "href": "/tmf-api/v5/policy/12345",
              "id": "12345",
              "name": "Data Retention Policy",
              "@referredType": "Policy",
              "version": "1.0"
            }
          ],
          "isEnabled": false,
          "id": "23456",
          "name": "Service Bundle"
        }
      ],
      "intent": {
        "@type": "IntentRef",
        "@schemaLocation": "/schema-registry/schema/IntentRef",
        "href": "/tmf-api/v5/intent/INT1001",
        "id": "INT1001",
        "name": "Upgrade Service",
        "@referredType": "Intent"
      },
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
          "owner": "XYZ Corp",
          "externalIdentifierType": "ProjectID",
          "id": "EXT45678"
        }
      ]
    },
    "state": "acknowledged"
  }
 ]
}
```


Update a resource order using RFC 7396 JSON Merge Patch for simple field updates

Request

```
PATCH /resourceOrder/EXNHG
Content-Type: application/merge-patch+json
{
  "@type": "ResourceOrder",
  "description": "Updated description: This order pertains to the deployment of a new
network infrastructure with enhanced security features.",
  "priority": 2,
  "expectedCompletionDate": "2025-02-20T06:41:23Z",
  "note": [
    {
```

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```
      "@type": "Note",
      "id": "NOTE002",
      "author": "Jane Smith",
      "date": "2025-01-25T10:30:00Z",
      "text": "Updated order priority and completion date due to additional security
requirements."
    }
  ]
}
```

Response

```
200
{
  "@type": "ResourceOrder",
  "href": "https://host:port/tmf-api/resourceOrdering/v5/resourceOrder/EXNHG",
  "id": "EXNHG",
  "@schemaLocation": "/schema-registry/schema/ResourceOrder",
  "creationDate": "2025-01-22T06:41:23Z",
  "cancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested cancellation due to project termination.",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "owner": "ExternalSystem",
      "externalIdentifierType": "ContractReference",
      "id": "EXT123456"
    }
  ],
  "name": "New Network Deployment Order",
  "description": "Updated description: This order pertains to the deployment of a new
network infrastructure with enhanced security features.",
  "priority": 2,
  "orderType": "Deployment",
  "category": "Infrastructure",
  "requestedStartDate": "2025-01-22T06:41:23Z",
  "requestedCompletionDate": "2025-02-15T06:41:23Z",
  "expectedCompletionDate": "2025-02-20T06:41:23Z",
  "note": [
    {
      "@type": "Note",
      "id": "NOTE002",
      "author": "Jane Smith",
      "date": "2025-01-25T10:30:00Z",
      "text": "Updated order priority and completion date due to additional security
requirements."
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/schema/RelatedParty",
      "role": "Service Provider",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/schema/Party",
        "href": "/tmf-api/v5/party/party123",
```

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```
        "id": "party123",
        "name": "Service Provider Organization",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "acknowledged",
  "orderItem": [
    {
      "@type": "ResourceOrderItem",
      "orderItemRelationship": [
        {
          "@type": "ResourceOrderItemRelationship",
          "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRelationship",
          "relationshipType": "dependency",
          "orderItem": {
            "@type": "ResourceOrderItemRef",
            "@schemaLocation": "/schema-registry/schema/ResourceOrderItemRef",
            "itemId": "ITEM56789",
            "resourceOrderId": "ORDER12345",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/ORDER12345",
            "@referredType": "ResourceOrderItem"
          }
        }
      ],
      "quantity": 1,
      "appointment": {
        "@type": "AppointmentRef",
        "@schemaLocation": "/schema-registry/schema/AppointmentRef",
        "href": "/tmf-api/v5/appointment/67890",
        "id": "67890",
        "name": "Site Installation Appointment",
        "@referredType": "InstallationAppointment",
        "description": "Appointment for the installation of network infrastructure at
the customer site."
      },
      "action": "add",
      "id": "12345",
      "resource": {
        "@type": "PhysicalResource",
        "@baseType": "Resource",
        "@schemaLocation": "/schema-registry/schema/Resource",
        "href": "/tmf-api/v5/resource/98765",
        "id": "98765",
        "category": "Network Element",
        "description": "Fiber Optic Cable connecting two locations for high-speed data
transmission.",
        "name": "Fiber Optic Cable - Main Line",
        "endOperatingDate": "2025-01-22T06:41:23Z",
        "administrativeState": "locked",
        "operationalState": "enabled",
        "usageState": "idle",
        "validFor": {
          "startDateTime": "2025-01-22T06:41:23Z",
          "endDateTime": "2025-02-25T06:41:23Z"
        },
        "note": [
          {
            "@type": "Note",
            "id": "12345",
```


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```
            "author": "John Doe",
            "date": "2025-01-22T06:41:23Z",
            "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
          }
        ],
        "resourceOrderItem": [
          {
            "@type": "RelatedResourceOrderItem",
            "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
            "@referredType": "ResourceOrderItem",
            "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
            "resourceOrderId": "RO12345",
            "itemAction": "add",
            "itemId": "ITEM67890",
            "role": "primary"
          }
        ],
        "place": [
          {
            "@type": "RelatedPlaceRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
            "role": "installation",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "/schema-registry/schema/PlaceRef",
              "href": "/tmf-api/v5/place/98765",
              "id": "98765",
              "name": "Customer Site - Location 1",
              "@referredType": "SiteLocation"
            }
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation": "/schema-registry/schema/RelatedPartyRefOrPartyRoleRef",
            "role": "Service Provider",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "href": "/tmf-api/v5/party/98765",
              "id": "98765",
              "name": "ABC Telecom Services",
              "@referredType": "TelecommunicationsProvider"
            }
          }
        ],
        "supportingResource": [
          {
            "@type": "PhysicalResource",
            "@baseType": "Resource",
            "@schemaLocation": "/schema-registry/schema/Resource",
            "href": "/tmf-api/v5/resource/54321",
            "id": "54321",
            "category": "Networking Equipment",
            "description": "Router used to manage network traffic between different
segments of the network.",
            "name": "Main Router - Data Center",
            "endOperatingDate": "2025-01-22T06:41:23Z",
```


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```
            "administrativeState": "locked",
            "operationalState": "enabled",
            "usageState": "idle",
            "validFor": {
              "startDateTime": "2025-01-22T06:41:23Z",
              "endDateTime": "2025-01-22T06:41:23Z"
            },
            "note": [
              {
                "@type": "Note",
                "id": "12345",
                "author": "John Doe",
                "date": "2025-01-22T06:41:23Z",
                "text": "This resource order was created to address a high-priority
customer request for infrastructure upgrade."
              }
            ],
            "resourceOrderItem": [
              {
                "@type": "RelatedResourceOrderItem",
                "@schemaLocation": "/schema-registry/schema/RelatedResourceOrderItem",
                "@referredType": "ResourceOrderItem",
                "resourceOrderHref": "/tmf-api/v5/resourceOrder/RO12345",
                "resourceOrderId": "RO12345",
                "itemAction": "add",
                "itemId": "ITEM67890",
                "role": "primary"
              }
            ],
            "place": [
              {
                "@type": "RelatedPlaceRef",
                "@schemaLocation": "/schema-registry/schema/RelatedPlaceRef",
                "role": "installation",
                "place": {
                  "@type": "PlaceRef",
                  "@schemaLocation": "/schema-registry/schema/PlaceRef",
                  "href": "/tmf-api/v5/place/98765",
                  "id": "98765",
                  "name": "Customer Site - Location 1",
                  "@referredType": "SiteLocation"
                }
              }
            ],
            "relatedParty": [
              {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@schemaLocation": "/schema-
registry/schema/RelatedPartyRefOrPartyRoleRef",
                "role": "Service Provider",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "@schemaLocation": "/schema-registry/schema/PartyRef",
                  "href": "/tmf-api/v5/party/98765",
                  "id": "98765",
                  "name": "ABC Telecom Services",
                  "@referredType": "TelecommunicationsProvider"
                }
              }
            ],
```


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```
            "resourceRelationship": [
              {
                "@type": "ResourceRelationship",
                "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
                "resourceRelationshipCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Performance Level",
                    "valueType": "Integer",
                    "value": 5,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "123",
                        "relationshipType": "isRelatedTo"
                      }
                    ]
                  }
                ],
                "resource": {
                  "@type": "ResourceRef",
                  "id": "56789",
                  "@referredType": "Resource",
                  "@schemaLocation": "/schema-registry/schema/Resource",
                  "href": "/tmf-api/v5/resource/56789",
                  "name": "Network Equipment"
                },
                "relationshipType": "dependsOn"
              }
            ],
            "resourceCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Processing Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "affects"
                  }
                ]
              }
            ],
            "attachment": [
              {
                "@type": "AttachmentRef",
                "@schemaLocation": "/schema-registry/schema/AttachmentRef",
                "href": "/tmf-api/v5/attachment/12345",
```


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```
                "id": "12345",
                "name": "Network Upgrade Plan",
                "@referredType": "TechnicalDocumentation",
                "description": "Detailed documentation for the upcoming network upgrade
project.",
                "url": "http://example.com/documents/network-upgrade"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "@schemaLocation": "/schema-registry/schema/ResourceSpecificationRef",
              "href": "/tmf-api/v5/specification/12345",
              "id": "12345",
              "name": "High-Speed Router",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "startOperatingDate": "2025-01-22T06:41:23Z",
            "resourceVersion": "2.0",
            "activationFeature": [
              {
                "@type": "Feature",
                "@schemaLocation": "/schema-registry/schema/Feature",
                "isBundle": true,
                "featureRelationship": [
                  {
                    "@type": "FeatureRelationship",
                    "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
                    "href": "/tmf-api/v5/feature/12345",
                    "id": "12345",
                    "name": "Advanced Security",
                    "relationshipType": "excluded",
                    "validFor": {
                      "startDateTime": "2025-01-22T06:41:23Z",
                      "endDateTime": "2025-01-22T06:41:23Z"
                    }
                  }
                ],
                "featureCharacteristic": [
                  {
                    "@type": "IntegerCharacteristic",
                    "@baseType": "Characteristic",
                    "@schemaLocation": "/schema-registry/schema/Characteristic",
                    "id": "12345",
                    "name": "Firewall Strength",
                    "valueType": "Integer",
                    "value": 10,
                    "characteristicRelationship": [
                      {
                        "@type": "CharacteristicRelationship",
                        "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                        "id": "12345",
                        "relationshipType": "enhances"
                      }
                    ]
                  }
                ],
                "policyConstraint": [
                  {
```


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```
                    "@type": "PolicyRef",
                    "@schemaLocation": "/schema-registry/schema/PolicyRef",
                    "href": "/tmf-api/v5/policy/12345",
                    "id": "12345",
                    "name": "Service Level Agreement",
                    "@referredType": "ServicePolicy",
                    "version": "1.0"
                  }
                ],
                "isEnabled": true,
                "id": "12345",
                "name": "Firewall Protection"
              }
            ],
            "intent": {
              "@type": "IntentRef",
              "@schemaLocation": "/schema-registry/schema/IntentRef",
              "href": "/tmf-api/v5/intent/12345",
              "id": "12345",
              "name": "Security Enhancement",
              "@referredType": "ServiceIntent"
            },
            "externalIdentifier": [
              {
                "@type": "ExternalIdentifier",
                "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
                "owner": "SecurityInc",
                "externalIdentifierType": "OrderReference",
                "id": "12345"
              }
            ]
          }
        ],
        "resourceRelationship": [
          {
            "@type": "ResourceRelationship",
            "@schemaLocation": "/schema-registry/schema/ResourceRelationship",
            "resourceRelationshipCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "12345",
                "name": "Bandwidth",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "12345",
                    "relationshipType": "dependent"
                  }
                ]
              }
            ],
            "resource": {
              "@type": "ResourceRef",
              "id": "87654",
```


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```
              "@referredType": "Resource",
              "@schemaLocation": "/schema-registry/schema/Resource",
              "href": "/tmf-api/v5/resource/87654",
              "name": "Server 01"
            },
            "relationshipType": "dependent"
          }
        ],
        "resourceCharacteristic": [
          {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "@schemaLocation": "/schema-registry/schema/Characteristic",
            "id": "CH123456",
            "name": "Storage Capacity",
            "valueType": "integer",
            "value": 1000,
            "characteristicRelationship": [
              {
                "@type": "CharacteristicRelationship",
                "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                "id": "CHREL7890",
                "relationshipType": "dependent"
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "AttachmentRef",
            "@schemaLocation": "/schema-registry/schema/Attachment",
            "href": "/tmf-api/v5/attachment/ATT98765",
            "id": "ATT98765",
            "name": "Product Specification",
            "@referredType": "Document",
            "description": "Detailed product specification document.",
            "url": "http://example.com/downloads/product-spec"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "@schemaLocation": "/schema-registry/schema/ResourceSpecification",
          "href": "/tmf-api/v5/specification/SPEC54321",
          "id": "SPEC54321",
          "name": "Enterprise Storage",
          "@referredType": "ResourceSpecification",
          "version": "2.1"
        },
        "startOperatingDate": "2025-01-22T06:41:23Z",
        "resourceVersion": "3.0",
        "activationFeature": [
          {
            "@type": "Feature",
            "@schemaLocation": "/schema-registry/schema/Feature",
            "isBundle": false,
            "featureRelationship": [
              {
                "@type": "FeatureRelationship",
                "@schemaLocation": "/schema-registry/schema/FeatureRelationship",
```


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```
                "href": "/tmf-api/v5/feature/12345",
                "id": "12345",
                "name": "Network Upgrade",
                "relationshipType": "excluded",
                "validFor": {
                  "startDateTime": "2025-01-22T06:41:23Z",
                  "endDateTime": "2025-01-22T06:41:23Z"
                }
              }
            ],
            "featureCharacteristic": [
              {
                "@type": "IntegerCharacteristic",
                "@baseType": "Characteristic",
                "@schemaLocation": "/schema-registry/schema/Characteristic",
                "id": "45678",
                "name": "Speed",
                "valueType": "Integer",
                "value": 1000,
                "characteristicRelationship": [
                  {
                    "@type": "CharacteristicRelationship",
                    "@schemaLocation": "/schema-
registry/schema/CharacteristicRelationship",
                    "id": "34567",
                    "relationshipType": "associated"
                  }
                ]
              }
            ],
            "policyConstraint": [
              {
                "@type": "PolicyRef",
                "@schemaLocation": "/schema-registry/schema/PolicyRef",
                "href": "/tmf-api/v5/policy/12345",
                "id": "12345",
                "name": "Data Retention Policy",
                "@referredType": "DataPolicy",
                "version": "1.0"
              }
            ],
            "isEnabled": false,
            "id": "23456",
            "name": "Service Bundle"
          }
        ],
        "intent": {
          "@type": "IntentRef",
          "@schemaLocation": "/schema-registry/schema/IntentRef",
          "href": "/tmf-api/v5/intent/INT1001",
          "id": "INT1001",
          "name": "Upgrade Service",
          "@referredType": "ServiceUpgrade"
        },
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "/schema-registry/schema/ExternalIdentifier",
            "owner": "XYZ Corp",
            "externalIdentifierType": "ProjectID",
```


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```
            "id": "EXT45678"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}
```

Deletes a ResourceOrder

`DELETE /resourceOrder/{id}`

Description

This operation deletes a ResourceOrder entity.

Usage samples

Delete a resource order by its unique identifier

Request

```
DELETE /resourceOrder/12345
Content-Type: application/json
```

Response

```
204
```

Operations on CancelResourceOrder

Retrieves a CancelResourceOrder by ID

`GET /cancelResourceOrder/{id}?fields=…`

Description

This operation retrieves a CancelResourceOrder entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a specific cancel resource order request using its unique identifier

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Request

```
GET /cancelResourceOrder/OSIQX
Content-Type: application/json
```

Response

```
200
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

List or find CancelResourceOrder objects

`GET /cancelResourceOrder?fields=…`

Description

This operation list CancelResourceOrder entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of cancel resource order requests with date range filtering and pagination

Request

```
GET /cancelResourceOrder?offset=10&limit=100&before=2025-01-21T00:00:00Z&after=2025-01-
01T00:00:00Z&sort=ascending
Content-Type: application/json
```

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Response

```
200
[
  {
    "@type": "CancelResourceOrder",
    "href": "https://host:port/tmf-api/cancelResourceOrder/54321",
    "id": "54321",
    "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
    "requestedCancellationDate": "2025-01-22T06:41:23Z",
    "effectiveCancellationDate": "2025-01-22T06:41:23Z",
    "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
    "state": "acknowledged",
    "resourceOrder": {
      "@type": "ResourceOrderRef",
      "@schemaLocation": "/schema-registry/schema/ResourceOrder",
      "href": "/tmf-api/v5/resourceOrder/12345",
      "id": "12345",
      "name": "Network Infrastructure Upgrade",
      "@referredType": "OrderReference"
    }
  }
]
```

Creates a CancelResourceOrder

`POST /cancelResourceOrder?fields=…`

Description

This operation creates a CancelResourceOrder entity.

Mandatory Attributes

Mandatory Attributes	Rule
resourceOrder	
resourceOrder.id	
@type	M

Usage samples

Create a new cancellation request for a resource order with immediate processing

Request

```
POST /cancelResourceOrder
Content-Type: application/json
```

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```
{
  "@type": "CancelResourceOrder",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Response

```
201
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Updates partially a CancelResourceOrder

`PATCH /cancelResourceOrder/{id}?fields=…`

Description

This operation allows partial updates of a CancelResourceOrder entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

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Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
cancellationReason	
effectiveCancellationDate	
href	
id	
requestedCancellationDate	
resourceOrder	
state	

Usage samples

Update a cancel resource order request using JSON Patch with query capabilities

Request

```
PATCH /cancelResourceOrder/DTEOL
Content-Type: application/json-patch-query+json
[
  {
    "op": "add",
    "path": "/account/settings",
    "value": "newSettingValue",
    "from": "system"
  },
  {
    "op": "add",
    "path": "/user/preferences",
    "value": "updatedPreference",
    "from": "userProfile"
  }
]
```

Response

```
200
```

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```
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Update a cancel resource order request using standard JSON merge patch

Request

```
PATCH /cancelResourceOrder/EHMIY
Content-Type: application/json
{
  "@type": "CancelResourceOrder",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Response

```
200
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
```

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```
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Update a cancel resource order request using RFC 6902 JSON Patch operations

Request

```
PATCH /cancelResourceOrder/ETUTY
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/account/settings",
    "value": "newSettingValue",
    "from": "system"
  },
  {
    "op": "add",
    "path": "/user/preferences",
    "value": "updatedPreference",
    "from": "userProfile"
  }
]
```

Response

```
200
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
```

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```
  }
}
```

Update a cancel resource order request using RFC 7396 JSON Merge Patch

Request

```
PATCH /cancelResourceOrder/SAKKL
Content-Type: application/merge-patch+json
{
  "@type": "CancelResourceOrder",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

Response

```
200
{
  "@type": "CancelResourceOrder",
  "href": "/tmf-api/v5/cancelResourceOrder/12345",
  "id": "12345",
  "@schemaLocation": "/schema-registry/schema/CancelResourceOrder",
  "requestedCancellationDate": "2025-01-22T06:41:23Z",
  "effectiveCancellationDate": "2025-01-22T06:41:23Z",
  "cancellationReason": "Customer requested to cancel due to a change in
requirements.",
  "state": "acknowledged",
  "resourceOrder": {
    "@type": "ResourceOrderRef",
    "@schemaLocation": "/schema-registry/schema/ResourceOrder",
    "href": "/tmf-api/v5/resourceOrder/12345",
    "id": "12345",
    "name": "Network Infrastructure Upgrade",
    "@referredType": "OrderReference"
  }
}
```

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Deletes a CancelResourceOrder

`DELETE /cancelResourceOrder/{id}`

Description

This operation deletes a CancelResourceOrder entity.

Usage samples

Delete a cancel resource order request by its unique identifier

Request

```
DELETE /cancelResourceOrder/12345
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
5.0.1	Jan'2025	Abhilash Prasad	V4 to V5 Migration, Updated Examples

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Version No	Date	Modified By	Description
TBD	Jan,2025	Abhilash Prasad, Jio	V4 to V5 Migration, Updated Examples

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